Query-Efficient Quantum Approximate Optimization via Graph-Conditioned Trust Regions

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(Dated:)

In low-depth implementations of the Quantum Approximate Optimization Algorithm (QAOA), the dominant cost is often the number of objective evaluations rather than circuit depth. We introduce a graph-conditioned trust-region method for reducing this query cost. A graph neural network predicts a Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ over QAOA angles. The mean initializes a local optimizer, the covariance defines an ellipsoidal trust region that constrains the search, and the predicted uncertainty determines an instance-dependent evaluation budget. Thus the learned distribution defines a search policy rather than only an initial parameter estimate.

Under explicit assumptions on local smoothness, curvature, calibration, and noise, we derive bounds on objective degradation within the trust region, lower bounds on gradient variance, preservation of expected objective ordering under depolarizing noise, and finite-sample coverage guarantees. We evaluate the method for MaxCut at depth $p = 2$ on Erdős–Rényi, 3-regular, Barabási–Albert, and Watts–Strogatz graphs with $n = 8$–16 vertices. Relative to random restarts and the strongest learned point-prediction baseline, the method reduces the mean number of circuit evaluations from 343 and 85 to 45 ± 7, while maintaining sampled approximation ratios within 3 percentage points of concentration-based heuristics.

The method does not improve absolute approximation ratios; its advantage is reduced query cost at comparable solution quality. The predictive uncertainty is calibrated in the experiments, with ECE$= 0.052$ and Spearman correlation $\rho = 0.770$, and the learned trust regions transfer to graph sizes not used during training. The results identify a low-depth, query-dominated regime in which graph-conditioned trust regions reduce the query cost of QAOA without modifying the ansatz.

I. INTRODUCTION

The Quantum Approximate Optimization Algorithm (QAOA) [1] is a widely studied variational algorithm for combinatorial optimization. In near-term settings, its cost is dominated by the classical outer loop, which repeatedly evaluates expectation values of the objective function. Even at low circuit depth, derivative-free optimization of the $2p$ variational parameters can require hundreds of objective evaluations per instance [2, 3]. On quantum hardware, each evaluation involves repeated state preparation and measurement, so the total number of circuit evaluations is a primary resource constraint [4, 5].

The question is whether a learned, graph-conditioned distribution over QAOA parameters can restrict search to an instance-dependent region and allocate computational effort across instances. We focus on low-depth QAOA, the regime most accessible to near-term devices and the regime in which the classical optimization loop can dominate total cost. The goal is to reduce objective evaluations, not to improve asymptotic approximation guarantees or to modify the QAOA ansatz.

Existing approaches to reducing the cost of parameter optimization include concentration-based schedules, learned initializations, and surrogate or Bayesian optimization methods. These approaches produce either fixed parameter choices or initial points for local optimization. In contrast, the learned Gaussian used here defines a search policy before any circuit evaluation is made: the mean initializes optimization, the covariance constrains the search trajectory, and the predicted uncertainty determines the evaluation budget. This differs from warm-start methods, which provide an initial point but do not constrain the subsequent optimization.

Given a graph $\mathcal{G}$, a Graph Isomorphism Network (GIN)-based predictor [6] with Laplacian spectral encodings [7, 8] outputs a Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}(\mathcal{G}), \boldsymbol{\Sigma}(\mathcal{G}))$ over QAOA angles. The mean initializes local optimization, the covariance defines a Mahalanobis trust region, and the trace of the covariance sets an instance-dependent evaluation budget. The learned model defines both the search geometry and the amount of computation assigned to each instance.

The analysis is restricted to low-depth QAOA and to the local behavior of the objective landscape near the predicted region. The results do not address global optimality or worst-case guarantees over arbitrary graph classes. They establish conditional guarantees under explicit assumptions on local smoothness, curvature, calibration, and noise.

The contributions are as follows.

1. **Graph-conditioned trust-region method.** The predicted Gaussian distribution is used to initialize optimization, constrain the search to an instance-dependent region, and allocate evaluation budgets. The method uses predictive uncertainty both to restrict the search region and to allocate query budget.

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2. **Conditional guarantees.** Under local regularity assumptions, we derive bounds on expected objective degradation within the trust region, lower bounds on gradient variance, preservation of expected objective ordering under depolarizing noise, and finite-sample coverage guarantees. These guarantees connect learned uncertainty to optimization geometry, robustness, and coverage.

3. **Empirical evaluation in the low-depth regime.** For MaxCut at depth p=2 on several graph families, the method reduces the number of circuit evaluations relative to both random initialization and learned point-prediction baselines, while maintaining comparable approximation ratios. The experiments are structured to isolate the contribution of distributional prediction beyond improved initialization alone.

We evaluate the method on Erdős–Rényi, 3-regular, Barabási–Albert, and Watts–Strogatz graph families with n=8–16 vertices. The learned trust region transfers to graph sizes not used during training and reduces query cost across all tested families. For n=14, the method reduces the mean number of circuit evaluations from 343 and 85 to 45 ± 7, while maintaining sampled approximation ratios within 3 percentage points of concentration-based heuristics.

The remainder of the paper is organized as follows. Section II defines the problem setting and notation. Section III discusses related work. Section IV introduces the predictor and trust-region method. Section V presents theoretical guarantees. Sections VI–VIII report empirical results and discuss limitations.

II. PROBLEM SETTING AND PRELIMINARIES

The setting is unweighted MaxCut and depth-p QAOA. The objective is the expected cost $F_{\mathcal{G}}(\boldsymbol{\theta})$, while the experiments also report the sampled best-bitstring approximation ratio obtained from a finite optimization run. The analysis concerns the behavior of $F_{\mathcal{G}}(\boldsymbol{\theta})$ near the predicted parameters at low depth.

A. MaxCut and the QAOA circuit

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an unweighted graph with $|\mathcal{V}| = n$ and $|\mathcal{E}| = m$. MaxCut seeks a bipartition $z \in \{+1, -1\}^n$ maximizing

$$C(z) = \sum_{(i,j)\in\mathcal{E}} \tfrac{1}{2}(1 - z_i z_j). \tag{1}$$

The problem is NP-hard [9]; the best known polynomial-time approximation ratio is the Goemans–Williamson bound $\alpha_{\mathrm{GW}} \approx 0.878$ [10], which is optimal under the Unique Games Conjecture [11, 12].

The depth-p QAOA prepares

$$|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle = e^{-i\beta_p B} e^{-i\gamma_p C} \cdots e^{-i\beta_1 B} e^{-i\gamma_1 C} |+\rangle^{\otimes n}, \tag{2}$$

where $C = \sum_{(i,j)\in\mathcal{E}} \frac{1}{2}(I - Z_i Z_j)$ is the cost Hamiltonian and $B = \sum_i X_i$ is the transverse-field mixer [1]. The variational parameters $\boldsymbol{\theta} = (\boldsymbol{\gamma}, \boldsymbol{\beta}) \in \mathbb{R}^{2p}$ are chosen to maximize the expected cost $F_{\mathcal{G}}(\boldsymbol{\theta}) = \langle \boldsymbol{\theta} | C | \boldsymbol{\theta} \rangle$.

We use two quality measures. The normalized expected objective $F_{\mathcal{G}}(\boldsymbol{\theta})/C_{\max}$ is used in the analysis. The sampled approximation ratio $r = C(z_{\text{best}})/C_{\max}$, where z_{best} is the best bitstring observed during optimization, is used in the experiments.

B. Angle concentration and landscape structure

For bounded-degree graph families at fixed depth p, optimal QAOA parameters exhibit concentration: the variance of optimal angles decreases as $O(1/n)$ [13, 14]. This structure enables cross-instance parameter transfer in both regular and heterogeneous graph families [3, 15]. The predictor introduced in Section IV exploits this phenomenon by placing the search in a region where high-quality parameters are likely to lie.

Variational quantum circuits can also exhibit barren plateaus, where the gradient variance at random parameters scales as

$$\mathrm{Var}_{\boldsymbol{\theta}}\big[\partial_j F(\boldsymbol{\theta})\big] \le O\big(2^{-n}\big), \tag{3}$$

for sufficiently deep or highly entangling ansätze [16–18]. For constant-depth QAOA with a local cost operator, this suppression is mitigated, but the landscape still contains extended regions of small gradient magnitude. Restricting optimization to a predicted region reduces exploration of such flat regions.

C. Spectral graph theory and positional encodings

The graph Laplacian $L = D - A$ admits the eigendecomposition $L = U \Lambda U^\top$, where eigenvectors $\{u_k\}_{k=0}^{n-1}$ (ordered by eigenvalue $0 = \lambda_0 \le \lambda_1 \le \cdots$) define spectral positional encodings [7, 8, 19]. The algebraic connectivity λ_2 (Fiedler value [20]) captures global connectivity and correlates with optimization difficulty.

Because eigenvectors are defined for arbitrary graph size, spectral encodings provide a size-invariant structural representation that supports cross-size transfer (section VII D). The sign ambiguity of eigenvectors is handled during training by random sign flips [8]. These encodings are combined with node-degree features and used as inputs to the graph-conditioned predictor.

D. Graph Isomorphism Networks

The Graph Isomorphism Network (GIN) [6] is as expressive as the Weisfeiler–Leman graph isomorphism test [21]. Each layer updates node representations according to

$$h_v^{(l+1)} = \mathrm{MLP}^{(l)}\big((1+\epsilon^{(l)})\,h_v^{(l)} + \textstyle\sum_{u\in\mathcal{N}(v)} h_u^{(l)}\big), \quad (4)$$

with learnable $\epsilon^{(l)}$ and neighbor set $\mathcal{N}(v)$. Graph-level representations are obtained by permutation-invariant pooling of node embeddings [22, 23]. This expressivity allows structural differences between graphs to be captured in the learned representation used for parameter prediction.

E. Wasserstein distance for Gaussians

Training compares predicted and target Gaussian distributions over parameters using the 2-Wasserstein distance. For probability measures P and Q on $\mathbb{R}^d$,

$$W_2(P,Q) = \Big(\inf_{\pi\in\Pi(P,Q)} \int \|\boldsymbol{x}-\boldsymbol{y}\|_2^2\, d\pi(\boldsymbol{x},\boldsymbol{y})\Big)^{1/2}. \quad (5)$$

For diagonal Gaussians this reduces to

$$W_2^2 = \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\|_2^2 + \|\boldsymbol{\sigma}_1 - \boldsymbol{\sigma}_2\|_2^2, \quad (6)$$

which can be evaluated and differentiated efficiently.

III. RELATION TO PRIOR WORK

Prior work relevant to this paper includes parameter transfer for QAOA, machine-learning approaches to combinatorial optimization, uncertainty quantification, and amortized optimization. The key distinction is the operational role of the learned model. The relevant questions are what is predicted (a fixed schedule, a point estimate, or a distribution), how that prediction is used during optimization (initialization only or a constraint on the subsequent search), and whether query effort is adapted across instances.

Parameter transfer for QAOA**.** Concentration of optimal angles [13, 14] has motivated several transfer strategies. Fixed-angle schedules [3] and interpolation-based warmstarts [2] exploit low-depth regularity to obtain strong approximation ratios on structured graph families without per-instance optimization. Extensions to heterogeneous graphs include cross-instance transfer [15], warm-starting from related circuits [24, 25], and graph-conditioned predictors based on GNNs [26] or reinforcement learning [27]. Additional approaches reduce query cost through Bayesian optimization [28], classical preprocessing [29], training without QPU access [30], unsupervised identification [31], and symmetry-aware search [32].

These methods provide fixed schedules, warm starts, or online search heuristics. The transferred information is used primarily to choose parameters before local optimization or to guide search as evaluations are gathered. Here the distribution is predicted before the first query. Its covariance restricts the admissible search region, and its calibrated uncertainty sets the evaluation budget.

Learned combinatorial heuristics. Machine learning is routinely used to guide combinatorial optimization [33–38]. In these settings, objective evaluations are inexpensive, and the primary metric is time to solution. In the variational quantum setting, each circuit evaluation dominates the cost, so the relevant quantity is the number of evaluations required to reach a given solution quality. The distinction shifts the emphasis from classical inference cost to query allocation. The paper therefore treats the learned model as part of the query policy, not merely as a fast heuristic for proposing a candidate solution.

Uncertainty quantification in scientific machine learning. Standard approaches include heteroscedastic regression [39, 40], deep ensembles [41], MC dropout [42], and calibrated prediction [43–46]. In the present setting, a single-network Gaussian parameterization is sufficient: the Wasserstein regularizer and contrastive loss structure the predicted variance without requiring ensemble methods. The resulting uncertainty directly determines the optimization strategy at test time. The role of uncertainty here is therefore prescriptive rather than merely diagnostic: it changes both the feasible region and the query budget.

Optimal transport and contrastive learning. The closed-form Wasserstein distance for Gaussian distributions [47] has been used in generative modeling [48, 49] and distributional robustness [50–52]. Here it serves as a regularizer for a distribution over variational parameters. Contrastive and metric-learning objectives [53–57] structure the representation so that the predicted distributions transfer across graph families.

Amortized optimization. Amortized inference replaces per-instance optimization with a learned mapping [58–61]. Existing amortized QAOA methods predict point initializations. In contrast, we predict a Gaussian distribution: the mean initializes optimization, the covariance constrains the search trajectory, and the uncertainty allocates the evaluation budget. The method remains hybrid rather than fully amortized: learning does not replace per-instance optimization, but reshapes it by restricting the search and reallocating expensive queries.

IV. GRAPH-CONDITIONED GAUSSIAN PREDICTION AND TRUST-REGION SEARCH

UQ-QAOA predicts a distribution over QAOA parameters and uses that distribution to structure subsequent optimization. The predictor specifies an initial point, a search region, and an evaluation budget. The method combines five components (fig. 2): a GIN-based predictor, Wasserstein-regularized training, contrastive metric learning, trust-region-constrained optimization, and uncertainty-guided budget allocation. The predictor output is used in three roles: the mean initializes optimization, the covariance defines the trust region, and the predicted uncertainty allocates evaluation budget.

Prediction and refinement are separated. A single forward pass produces $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, after which optimization is restricted to a region and budget determined by the prediction. The learned distribution therefore determines the optimization policy, not only its initialization.

a. Standing assumptions. The guarantees in section V are local. They assume: (A1) $F_{\mathcal{G}}$ is Lipschitz-continuous on the trust region (Proposition 1); (A2) $\lambda_{\min}(-\nabla^2 F_{\mathcal{G}}(\boldsymbol{\mu})) \geq \kappa > 0$ (Theorem 4); and (A3) depolarizing noise with per-layer strength $\varepsilon < 1$ (Theorem 16). Assumption (A2) concerns the neighborhood reached by the predictor and does not assert global structure of the QAOA landscape.

A. Gaussian predictor architecture

Given $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, node features are formed by concatenating normalized degree d_v/n with k-dimensional Laplacian spectral positional encodings [7, 8, 19], i.e., the k smallest nontrivial eigenvectors of $L = D - A$, yielding $h_v^{(0)} \in \mathbb{R}^{k+1}$. Eigenvector sign ambiguity is handled by random sign flips during training [8].

A three-layer GIN with hidden dimension d=64 and LayerNorm [62] produces node embeddings. A graph-level representation $\boldsymbol{g} \in \mathbb{R}^{2d}$ is obtained by concatenating mean and max pooling:

$$\boldsymbol{g} = \left[\tfrac{1}{n}\textstyle\sum_v h_v^{(L)} \;\middle\|\; \max_v h_v^{(L)}\right]. \tag{7}$$

Two linear heads produce the Gaussian parameters:

$$\boldsymbol{\mu} = W_\mu \boldsymbol{g} + b_\mu \in \mathbb{R}^{2p}, \tag{8}$$

$$\log \boldsymbol{\sigma}^2 = \mathrm{clamp}(W_\sigma \boldsymbol{g} + b_\sigma, -5, 2). \tag{9}$$

The predictive distribution is

$$p(\boldsymbol{\theta} \mid \mathcal{G}) = \mathcal{N}\big(\boldsymbol{\mu}(\mathcal{G}),\, \mathrm{diag}(\boldsymbol{\sigma}^2(\mathcal{G}))\big).$$

The clamp in eq. (9) ensures numerical stability with $\sigma_j \in [e^{-5/2}, e]$. We restrict to diagonal covariance. At p=2 this gives axis-aligned trust regions and is adequate for the regime studied here. It does not model correlations between angles. At larger depth, full-covariance, low-rank, mixture, or torus-aware models may be needed.

B. Wasserstein-regularized training

Data. Training uses 240 graphs at n=14 (60 per family) and a validation set of 80 graphs. Targets $\boldsymbol{\theta}_i^*$ are obtained by multi-restart Nelder–Mead [63] on exact simulation. These are local optima (see appendix A) and define the regions used for transfer.

Phase 1.

$$\mathcal{L}_1 = \frac{1}{N}\sum_{i=1}^{N} \|\boldsymbol{\mu}(\mathcal{G}_i) - \boldsymbol{\theta}_i^*\|_2^2 + \lambda_C \mathcal{L}_{\mathrm{CL}}. \tag{10}$$

Phase 2.

$$\mathcal{L}_2 = \mathcal{L}_{\mathrm{NLL}} + \lambda_W \mathcal{L}_{\mathrm{W2}} + \lambda_C \mathcal{L}_{\mathrm{CL}}, \tag{11}$$

The negative log-likelihood is

$$\mathcal{L}_{\mathrm{NLL}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{2p}\left[\frac{(\mu_j(\mathcal{G}_i) - \theta_{ij}^*)^2}{\sigma_j^2(\mathcal{G}_i)} + \log \sigma_j^2(\mathcal{G}_i)\right]. \tag{12}$$

The phased schedule prevents trivial variance inflation that can occur when optimizing NLL from initialization.

Wasserstein regularizer. For diagonal Gaussians,

$$W_2^2(P_i, P_j) = \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|_2^2 + \|\boldsymbol{\sigma}_i - \boldsymbol{\sigma}_j\|_2^2.$$

Weights $w_{ij} = \exp(-\|\boldsymbol{\theta}_i^* - \boldsymbol{\theta}_j^*\|_2^2 / 2\tau_W^2)$ encode angle-space similarity:

$$\mathcal{L}_{\mathrm{W2}} = \frac{1}{|\mathcal{B}|^2}\sum_{i \neq j \in \mathcal{B}} w_{ij}\, W_2^2(P_i, P_j). \tag{13}$$

C. Contrastive metric learning

A supervised contrastive loss structures the embedding $\boldsymbol{g}$. For anchor i, define

$$\mathcal{P}(i) = \big\{ j \neq i : \|\boldsymbol{\theta}_i^* - \boldsymbol{\theta}_j^*\|_2 < \delta \big\}.$$

$$\mathcal{L}_{\mathrm{CL}} = -\frac{1}{|\mathcal{B}|}\sum_i \frac{1}{|\mathcal{P}(i)|}\sum_{j \in \mathcal{P}(i)} \log \frac{e^{\mathrm{sim}(\boldsymbol{g}_i, \boldsymbol{g}_j)/\tau_C}}{\sum_{k \neq i} e^{\mathrm{sim}(\boldsymbol{g}_i, \boldsymbol{g}_k)/\tau_C}}, \tag{14}$$

The contrastive loss maps graphs with similar QAOA landscapes to nearby representations, improving transfer.

D. Trust-region QAOA optimization

The predicted covariance defines a constraint on the search.

Definition 1 (Trust region)**.** *Given a predicted Gaussian $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for graph $\mathcal{G}$, with $\boldsymbol{\Sigma}$ symmetric positive definite, and a confidence level $\alpha \in (0,1)$, the* trust region *is the Mahalanobis ellipsoid*

$$\mathcal{T}_\alpha(\mathcal{G}) = \big\{ \boldsymbol{\theta} \in \mathbb{R}^{2p} : (\boldsymbol{\theta} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}) \leq \chi^2_{2p}(\alpha) \big\}, \tag{15}$$

where $\chi^2_{2p}(\alpha)$ is the α-quantile of the χ^2 distribution with $2p$ degrees of freedom.

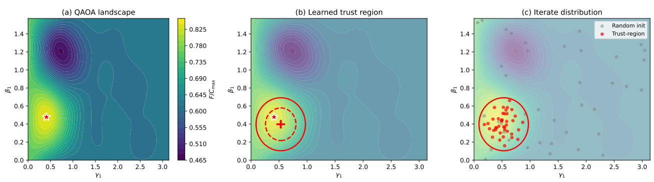

FIG. 1. Trust-region mechanism on an n=8 Erdős–Rényi instance at p=2 (layer-1 slice). (a) Landscape of the QAOA objective, with the optimum marked by a star. (b) Learned Mahalanobis trust regions at 95% (solid) and 68% (dashed), concentrated around the high-quality basin. (c) Random initializations (gray) versus trust-region samples (red), showing that the learned ellipsoid concentrates evaluations near the optimum.

Optimization is restricted to $\mathcal{T}_\alpha$: (a) candidate angles $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ are sampled from the corresponding truncated Gaussian restricted to $\mathcal{T}_\alpha$; (b) during Nelder–Mead refinement, any simplex vertex $\boldsymbol{\theta}'$ exiting $\mathcal{T}_\alpha$ is projected back:

$$\mathrm{proj}_{\mathcal{T}}(\boldsymbol{\theta}') = \boldsymbol{\mu} + \min\Bigl(1, \frac{\sqrt{\chi^2_{2p}(\alpha)}}{\|\boldsymbol{\Sigma}^{-1/2}(\boldsymbol{\theta}' - \boldsymbol{\mu})\|_2}\Bigr)(\boldsymbol{\theta}' - \boldsymbol{\mu}). \tag{16}$$

This constrains the entire optimization trajectory, rather than only its initialization [24, 25].

Figure 1 makes the trust-region idea concrete on an n=8 Erdős–Rényi instance at p=2. The predicted covariance localizes the search near a high-quality basin of the landscape. The covariance therefore acts as a geometric prior, not only as an uncertainty estimate.

E. Uncertainty-guided adaptive refinement

Define

$$U(\mathcal{G}) = \mathrm{tr}(\boldsymbol{\Sigma}(\mathcal{G}))/2p, \quad z(\mathcal{G}) = \frac{U(\mathcal{G}) - U_{\mathrm{med}}}{U_{\mathrm{iqr}}}.$$

Resource allocation:

$$K(\mathcal{G}) = \mathrm{clamp}(\lfloor 1 + 4\sigma(z)\rfloor, 1, 5), \tag{17}$$
$$T(\mathcal{G}) = \mathrm{clamp}(\lfloor T_{\mathrm{base}}(0.5 + z_+)\rfloor, 5, 2T_{\mathrm{base}}), \tag{18}$$

Higher uncertainty increases both the number of samples and the number of refinement steps. The mapping is monotone and bounded, enabling comparison with fixed-budget baselines.

Figure 2 reinforces the conceptual separation between prediction and search. The learned model is used once to produce a Gaussian, and that Gaussian then determines the local optimization policy: where the search begins, how far it is allowed to move, and how much computation the instance receives.

Remark 1 (Conformal trust-region radius). *At inference time, the $\chi^2_{2p}(\alpha)$ radius in Step 5 can be replaced by the conformal quantile $\hat{q}_{1-\alpha}$ (proposition 14), yielding a* conformalized *trust region*

$$\mathcal{T}_{\hat{q}_{1-\alpha}} = \bigl\{\boldsymbol{\theta} : (\boldsymbol{\theta} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}) \le \hat{q}_{1-\alpha}\bigr\}$$

with the distribution-free, finite-sample guarantee $\mathbb{P}[\boldsymbol{\theta}^* \in \mathcal{T}_{\hat{q}_{1-\alpha}}] \ge 1 - \alpha$. For our validation set ($\hat{q}_{0.90} = 10.8$ vs. $\chi^2_4(0.90) = 7.779$), this inflates the trust region by a factor of $(10.8/7.779)^{1/2} \approx 1.178$ in radius, i.e., an approximately 18% enlargement. This trades a larger search region for a distribution-free finite-sample marginal coverage statement.

Algorithm 1: UQ-QAOA **inference**

Input: graph $\mathcal{G}$, trained model $f_\theta : \mathcal{G} \mapsto (\boldsymbol{\mu}, \boldsymbol{\Sigma})$, budget T_{base}, confidence level α.

Output: approximate MaxCut bitstring z^*.

1. Predict $(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \leftarrow f_\theta(\mathcal{G})$.
2. Compute $U \leftarrow \mathrm{tr}(\boldsymbol{\Sigma})/2p$; $z \leftarrow (U - U_{\mathrm{med}})/U_{\mathrm{iqr}}$.
3. Set sample count $K \leftarrow \mathrm{clamp}(\lfloor 1 + 4\sigma(z)\rfloor, 1, 5)$.
4. Set polish budget $T \leftarrow \mathrm{clamp}(\lfloor T_{\mathrm{base}}(0.5 + z_+)\rfloor, 5, 2T_{\mathrm{base}})$.
5. Construct trust region $\mathcal{T}_\alpha \leftarrow \{\boldsymbol{\theta} : \|\boldsymbol{\Sigma}^{-1/2}(\boldsymbol{\theta} - \boldsymbol{\mu})\|_2^2 \le \chi^2_{2p}(\alpha)\}$.
6. Sample $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K$ from the truncated Gaussian $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ restricted to $\mathcal{T}_\alpha$.
7. Evaluate QAOA circuit: $\boldsymbol{\theta}^* \leftarrow \arg\max_k F_{\mathcal{G}}(\boldsymbol{\theta}_k)$.
8. Trust-region polish: run T iterations of Nelder–Mead from $\boldsymbol{\theta}^*$, projecting each iterate via eq. (16).
9. Return best bitstring z^* from final QAOA measurement.

FIG. 2. UQ-QAOA inference. The predictor first converts uncertainty into an instance-dependent sampling and refinement budget (Steps 3–4), then performs trust-region-constrained optimization (Steps 5–8). The total number of circuit evaluations is $K + T$, up to constant factors from the refinement routine.

V. GUARANTEES FOR LEARNED TRUST-REGION SEARCH

The following results give guarantees for the learned distribution and the induced trust-region search. The results are local and conditional. They do not assert global optimality of the hybrid procedure. They give conditions under which the predicted region preserves objective quality, supports coverage statements, and justifies uncertainty-based resource allocation.

Throughout, $\mathcal{G} \sim \mathcal{D}$ denotes a held-out graph except where otherwise stated. The predictor is $f_\theta(\mathcal{G}) = (\boldsymbol{\mu}(\mathcal{G}), \boldsymbol{\Sigma}(\mathcal{G}))$. The reference point $\boldsymbol{\theta}^*(\mathcal{G})$ denotes the local optimum produced by the multi-restart Nelder–Mead procedure used for target generation, or the optimum specified in the corresponding statement. The trust region $\mathcal{T}_\alpha(\mathcal{G})$ is defined in eq. (15). When the radius is not tied to a $\chi^2_{2p}(\alpha)$ quantile, we write $\mathcal{T}_q(\mathcal{G})$.

Expectations and probabilities are taken with respect to the randomness explicitly indicated: draws of $\mathcal{G}$ from

$\mathcal{D}$, samples from $\mathcal{N}(\boldsymbol{\mu}(\mathcal{G}), \boldsymbol{\Sigma}(\mathcal{G}))$, or uniform samples on $\mathcal{T}_\alpha(\mathcal{G})$. Constants such as $L_{\mathcal{G},K}$, $L_\mathcal{G}$, κ, Λ, and M_3 are deterministic once $\mathcal{G}$ is fixed.

A. Landscape properties

Proposition 1 (Local Lipschitz regularity of the QAOA objective)**.** *For a connected graph $\mathcal{G}$ with m edges, the depth-p QAOA objective $F_\mathcal{G} : \mathbb{R}^{2p} \to \mathbb{R}$ is real analytic. In particular, on any bounded set $K \subset \mathbb{R}^{2p}$,*

$$L_{\mathcal{G},K} := \sup_{\boldsymbol{\theta}\in K} \|\nabla F_\mathcal{G}(\boldsymbol{\theta})\|_2 < \infty. \tag{19}$$

A conservative global bound is $L_\mathcal{G} \le 2m\sqrt{2p}$.

Proof. Analyticity follows because $U_{\mathrm{QAOA}}(\boldsymbol{\theta})$ is a finite product of matrix exponentials. Thus $F_\mathcal{G}$ is analytic and $\nabla F_\mathcal{G}$ is continuous. On compact K, the supremum is finite, and the mean value theorem yields the Lipschitz bound. The global estimate follows from standard derivative bounds for QAOA expectation values [4, 64]. □

B. Trust-region guarantees

Theorem 2 (Lower bound on the expected objective within the trust region)**.** *Let $\mathcal{T}_\alpha(\mathcal{G})$ be defined by eq. (15). Suppose $F_\mathcal{G}(\boldsymbol{\mu}) \ge r^* C_{\max}$. Let Θ be supported on $\mathcal{T}_\alpha(\mathcal{G})$. Then*

$$\mathbb{E}\big[F_\mathcal{G}(\Theta)\big] \ge r^* C_{\max} - L_\mathcal{G}\sqrt{\chi^2_{2p}(\alpha)\cdot\|\boldsymbol{\Sigma}\|_{\mathrm{op}}}, \tag{20}$$

Proof. For $\boldsymbol{\theta} \in \mathcal{T}_\alpha$, one has

$$\|\boldsymbol{\theta}-\boldsymbol{\mu}\|_2^2 \le \|\boldsymbol{\Sigma}\|_{\mathrm{op}}\cdot(\boldsymbol{\theta}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\boldsymbol{\theta}-\boldsymbol{\mu}) \le \|\boldsymbol{\Sigma}\|_{\mathrm{op}}\cdot\chi^2_{2p}(\alpha).$$

Hence

$$\|\boldsymbol{\theta}-\boldsymbol{\mu}\|_2 \le \sqrt{\chi^2_{2p}(\alpha)\|\boldsymbol{\Sigma}\|_{\mathrm{op}}}.$$

By Proposition 1,

$$|F_\mathcal{G}(\boldsymbol{\theta}) - F_\mathcal{G}(\boldsymbol{\mu})| \le L_\mathcal{G}\sqrt{\chi^2_{2p}(\alpha)\|\boldsymbol{\Sigma}\|_{\mathrm{op}}}.$$

Applying the assumption $F_\mathcal{G}(\boldsymbol{\mu}) \ge r^* C_{\max}$ and taking expectation yields the claim. □

Corollary 3 (Expected objective under variance confinement)**.** *Under the conditions of Theorem 2, if $\sigma^2_{\max} \le (r^* C_{\max})^2/(4L^2_\mathcal{G}\chi^2_{2p}(\alpha))$, then $\mathbb{E}[F_\mathcal{G}(\Theta)] \ge \frac{1}{2}r^* C_{\max}$.*

Proof. Set $\epsilon = \frac{1}{2}r^* C_{\max}$ in Theorem 2. This gives the sufficient condition $\sigma^2_{\max} \le (r^* C_{\max})^2/(4L^2_\mathcal{G}\chi^2_{2p}(\alpha))$. The conclusion follows immediately. □

C. Gradient anti-concentration

Theorem 4 (Local gradient anti-concentration in the trust region)**.** *Assume $F_\mathcal{G}$ is three times continuously differentiable near $\mathcal{T}_\alpha(\mathcal{G})$, $\lambda_{\min}(-\nabla^2 F_\mathcal{G}(\boldsymbol{\mu})) \ge \kappa > 0$, and third derivatives are bounded by M_3. Then for $\boldsymbol{\theta} \sim \mathrm{Unif}(\mathcal{T}_\alpha)$,*

$$\begin{aligned}\mathrm{Var}_{\boldsymbol{\theta}\sim\mathrm{Unif}(\mathcal{T}_\alpha)}\big[\partial_j F_\mathcal{G}(\boldsymbol{\theta})\big] \ge{}& \frac{\kappa^2\,\sigma^2_{\min}\,\chi^2_{2p}(\alpha)}{2p+2} \\ &- C_{p,\alpha}\big(\Lambda M_3\sigma^3_{\max} + M_3^2\sigma^4_{\max}\big),\end{aligned} \tag{21}$$

where $\Lambda := \|\nabla^2 F_\mathcal{G}(\boldsymbol{\mu})\|_{\mathrm{op}}$, $\sigma_{\min} = \min_j \sigma_j$, and $\sigma_{\max} = \max_j \sigma_j$.

Proof. See appendix H. □

D. Algorithmic guarantees

Proposition 5 (Feasibility and monotonic best-so-far value under projection)**.** *Projected optimization within $\mathcal{T}_\alpha$ produces feasible iterates and a nondecreasing best-so-far objective. The final iterate dominates the initial sampled candidates. Let $\{\boldsymbol{\theta}_t\}_{t=0}^T$ be the feasible iterates generated by projected Nelder–Mead within $\mathcal{T}_\alpha$, define $b_t := \max_{0\le s\le t} F_\mathcal{G}(\boldsymbol{\theta}_s)$, and let $\widehat{\boldsymbol{\theta}}_t$ denote any feasible iterate attaining b_t. Then:*

(i) $\boldsymbol{\theta}_t \in \mathcal{T}_\alpha$ for all t.

(ii) b_t is non-decreasing in t.

(iii) $F_\mathcal{G}(\widehat{\boldsymbol{\theta}}_T) \ge \max_k F_\mathcal{G}(\boldsymbol{\theta}_k)$ where $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K$ are the initial feasible samples used to seed the search.

Proof. (i) The projection operator eq. (16) maps every point to $\mathcal{T}_\alpha$: if $\boldsymbol{\theta}' \in \mathcal{T}_\alpha$, the projection is the identity; otherwise, it rescales $\boldsymbol{\theta}' - \boldsymbol{\mu}$ to the boundary of $\mathcal{T}_\alpha$. (ii) Monotonicity is immediate from the definition $b_t = \max_{0\le s\le t} F_\mathcal{G}(\boldsymbol{\theta}_s)$: the index set on which the maximum is taken only enlarges with t. (iii) Because the initial sampled points are among the feasible points seen by the algorithm before refinement begins, $b_T \ge \max_{1\le k\le K} F_\mathcal{G}(\boldsymbol{\theta}_k)$. Since $\widehat{\boldsymbol{\theta}}_T$ attains b_T, the claim follows. □

Corollary 6 (Output guarantee under trust-region capture)**.** *Let $\boldsymbol{\theta}_{\mathrm{out}}$ denote the best feasible angle vector returned by Algorithm 2, and let z_{out} be the final bitstring sampled from the QAOA circuit at $\boldsymbol{\theta}_{\mathrm{out}}$. Fix a trust region $\mathcal{T}_q(\mathcal{G})$ and assume the reference local optimum $\boldsymbol{\theta}^*(\mathcal{G})$ belongs to that region. Then*

$$F_\mathcal{G}(\boldsymbol{\theta}_{\mathrm{out}}) \ge F_\mathcal{G}\big(\boldsymbol{\theta}^*(\mathcal{G})\big) - 2L_\mathcal{G}\sqrt{q\,\|\boldsymbol{\Sigma}(\mathcal{G})\|_{\mathrm{op}}}, \tag{22}$$

and therefore

$$\begin{aligned}\mathbb{E}\big[C(z_{\mathrm{out}}) \mid \boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_q(\mathcal{G})\big] \ge{}& F_\mathcal{G}\big(\boldsymbol{\theta}^*(\mathcal{G})\big) \\ &- 2L_\mathcal{G}\sqrt{q\,\|\boldsymbol{\Sigma}(\mathcal{G})\|_{\mathrm{op}}}.\end{aligned} \tag{23}$$

If, in addition, the trust region has capture probability $p_{\mathrm{cap}}(\mathcal{G})$, then since MaxCut costs are nonnegative,

$$\mathbb{E}\big[C(z_{\mathrm{out}})\big] \geq p_{\mathrm{cap}}(\mathcal{G})\Big[F_{\mathcal{G}}\big(\boldsymbol{\theta}^*(\mathcal{G})\big) - 2L_{\mathcal{G}}\sqrt{q\,\|\boldsymbol{\Sigma}(\mathcal{G})\|_{\mathrm{op}}}\Big]. \tag{24}$$

For the algorithmic χ^2 region one may take $p_{\mathrm{cap}} \geq \alpha - \eta$ from Proposition 15; for the conformalized region one has $p_{\mathrm{cap}} \geq 1 - \alpha$.

Proof. Under the capture event, every sampled candidate and every projected iterate lies in $\mathcal{T}_q(\mathcal{G})$. Proposition 15 therefore gives the pointwise bound

$$F_{\mathcal{G}}(\boldsymbol{\theta}) \geq F_{\mathcal{G}}\big(\boldsymbol{\theta}^*(\mathcal{G})\big) - 2L_{\mathcal{G}}\sqrt{q\,\|\boldsymbol{\Sigma}(\mathcal{G})\|_{\mathrm{op}}} \qquad \forall\,\boldsymbol{\theta} \in \mathcal{T}_q(\mathcal{G}).$$

Since Algorithm 2 returns an angle vector inside that region, eq. (22) follows. By definition of the QAOA objective, $\mathbb{E}[C(z_{\mathrm{out}}) \mid \boldsymbol{\theta}_{\mathrm{out}}] = F_{\mathcal{G}}(\boldsymbol{\theta}_{\mathrm{out}})$, which gives eq. (23). The unconditional bound eq. (24) follows by conditioning on the capture event and using $C(z_{\mathrm{out}}) \geq 0$ for MaxCut. □

E. Sampling guarantees

Theorem 7 (Best-of-K guarantee). *Let $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K \overset{i.i.d.}{\sim} \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Then*

$$\mathbb{E}\big[\max_k F_{\mathcal{G}}(\boldsymbol{\theta}_k)\big] \geq F_{\mathcal{G}}(\boldsymbol{\mu}) - \frac{L_{\mathcal{G}}\sqrt{\mathrm{tr}(\boldsymbol{\Sigma})}}{\sqrt{K}}. \tag{25}$$

Proof. See appendix C. □

Lemma 8 (Trust-region volume ratio). *For a diagonal covariance $\boldsymbol{\Sigma} = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_{2p}^2)$ with confidence α=0.95 and p=2, the trust-region volume relative to the unconstrained search hypercube $[-\pi, \pi]^{2p}$ is*

$$\frac{\mathrm{Vol}(\mathcal{T}_{0.95})}{\mathrm{Vol}([-\pi, \pi]^{2p})} = \frac{\pi^p[\chi_4^2(0.95)]^p}{\Gamma(p+1)\,(2\pi)^{2p}} \prod_{j=1}^{4} \sigma_j. \tag{26}$$

If one uses the isotropic proxy $\sigma_j \equiv \bar{\sigma} = 0.15$, then eq. (26) gives $0.2850\,\bar{\sigma}^4 \approx 1.44 \times 10^{-4}$.

Proof. The trust region is a 4-dimensional ellipsoid with half-lengths $\sigma_j\sqrt{\chi_4^2(0.95)}$. Its volume is $\frac{\pi^2}{2}[\chi_4^2(0.95)]^2 \prod_j \sigma_j$, and the hypercube volume is $(2\pi)^4$. The ratio follows by direct computation with $\chi_4^2(0.95) \approx 9.49$. □

F. Generalization and calibration

The remaining results address out-of-sample transfer and calibration of the learned distribution.

Proposition 9 (Wasserstein continuity in a WL-compatible pseudometric). *If the* GIN *encoder ϕ is L_{enc}-Lipschitz with respect to a WL-compatible graph pseudometric d_{WL} and the prediction heads are L_{head}-Lipschitz, then:*

$$W_2\big(f_\theta(\mathcal{G}), f_\theta(\mathcal{G}')\big) \leq \sqrt{2}\,L_{\mathrm{head}}L_{\mathrm{enc}}\,d_{\mathrm{WL}}(\mathcal{G}, \mathcal{G}'), \tag{27}$$

where $d_{\mathrm{WL}}(\mathcal{G}, \mathcal{G}') = 0$ is allowed for non-isomorphic graphs that are indistinguishable by the relevant WL statistics.

Proof. See appendix E. □

Theorem 10 (Adaptive-allocation guarantee under calibration and monotone difficulty). *Assume approximate calibration and the regret envelope eq. (29). Suppose the predicted uncertainty satisfies η-approximate calibration:*

$$\big|\mathbb{P}[\|\boldsymbol{\theta}^* - \boldsymbol{\mu}\|_{\boldsymbol{\Sigma}^{-1}} \leq r] - F_{\chi^2_{2p}}(r^2)\big| \leq \eta \quad \forall\, r > 0, \tag{28}$$

where $F_{\chi^2_{2p}}$ is the χ^2_{2p} CDF. Assume the residual optimization regret satisfies the instance-wise envelope

$$R_\pi(\mathcal{G}) \leq c_0 + c_1\frac{\sqrt{U(\mathcal{G}) + \eta}}{\sqrt{K_\pi(\mathcal{G})}} \tag{29}$$

where $U(\mathcal{G}) = \mathrm{tr}(\boldsymbol{\Sigma}(\mathcal{G}))/2p$ and $c_0, c_1 \geq 0$ are independent of π. Define population regret by

$$\mathrm{Regret}(\pi) := \mathbb{E}_{\mathcal{G}\sim\mathcal{D}}[R_\pi(\mathcal{G})]. \tag{30}$$

Let $\bar{K} := \mathbb{E}_{\mathcal{D}}[K_{\mathrm{adapt}}(\mathcal{G})]$, and let π_{unif} denote the mean-budget-matched uniform comparator with $K_{\mathrm{unif}}(\mathcal{G}) \equiv \bar{K}$. Assume further that the adaptive allocation rule satisfies the envelope-dominance condition

$$\mathbb{E}_{\mathcal{D}}\left[\frac{\sqrt{U(\mathcal{G}) + \eta}}{\sqrt{K_{\mathrm{adapt}}(\mathcal{G})}}\right] \leq \frac{1}{\sqrt{\bar{K}}}\mathbb{E}_{\mathcal{D}}\left[\sqrt{U(\mathcal{G}) + \eta}\right]. \tag{31}$$

Then the adaptive allocation strategy π_{adapt} (Eqs. 17–18) satisfies

$$\begin{aligned} \mathrm{Regret}(\pi_{\mathrm{adapt}}) &\leq c_0 + c_1\,\mathbb{E}_{\mathcal{D}}\left[\frac{\sqrt{U(\mathcal{G}) + \eta}}{\sqrt{K_{\mathrm{adapt}}(\mathcal{G})}}\right] \\ &\leq c_0 + \frac{c_1}{\sqrt{\bar{K}}}\mathbb{E}_{\mathcal{D}}\left[\sqrt{U(\mathcal{G}) + \eta}\right], \qquad (32) \\ \mathbb{E}[B_{\pi_{\mathrm{adapt}}}] &= \alpha\,\mathbb{E}[B_{\pi_{\mathrm{unif}}}], \qquad (33) \end{aligned}$$

where B_π is the total evaluation budget under policy π and $\alpha = \mathbb{E}_{\mathcal{D}}[B_{\pi_{\mathrm{adapt}}}]/\mathbb{E}_{\mathcal{D}}[B_{\pi_{\mathrm{unif}}}]$ depends on the deployment distribution of predicted uncertainties. Thus $\alpha < 1$ is not a consequence of calibration alone; it depends on the deployment distribution together with the implemented allocation rule.

Proof. See appendix D. □

Proposition 11 (Monotonicity of the implemented allocation rule)**.** *Let $z(\mathcal{G}) = (U(\mathcal{G}) - U_{\text{med}})/U_{\text{iqr}}$, and define K_{adapt} and T_{adapt} by eqs. (17) and (18). Then both allocations are nondecreasing functions of U, and one has the explicit thresholds*

$$K_{\text{adapt}}(\mathcal{G}) = \begin{cases} 1, & z(\mathcal{G}) < -\log 3, \\ 2, & -\log 3 \le z(\mathcal{G}) < 0, \\ 3, & 0 \le z(\mathcal{G}) < \log 3, \\ 4, & z(\mathcal{G}) \ge \log 3. \end{cases} \tag{34}$$

If $T_{\text{base}} \ge 10$, then $T_{\text{adapt}}(\mathcal{G}) = \lfloor T_{\text{base}}/2 \rfloor$ for $z(\mathcal{G}) \le 0$, and T_{adapt} increases linearly in $z(\mathcal{G})$ until it reaches the cap $2T_{\text{base}}$. Thus the implemented policy reallocates computation monotonically from the low-U region to the high-U region, which is the ordering premise behind eq. (31). The same ordering is supported empirically by fig. 9 and table I: larger predicted uncertainty is associated with harder instances, and exploiting that ordering lowers the total query count.

Proof. The sigmoid is strictly increasing, so $\lfloor 1+4\sigma(z) \rfloor$ is nondecreasing in z. The thresholds follow from the identities $\sigma(z) = 1/4$ at $z = -\log 3$, $\sigma(z) = 1/2$ at $z = 0$, and $\sigma(z) = 3/4$ at $z = \log 3$. Because $\sigma(z) < 1$ for finite z, the implemented rule takes the values $1, 2, 3, 4$ on the four regimes in eq. (34). The monotonicity of T_{adapt} follows in the same way from the monotonicity of $z_+ = \max(z, 0)$ and the clamp map. If $z \le 0$, then $z_+ = 0$ and $T_{\text{adapt}} = \text{clamp}(\lfloor T_{\text{base}}/2 \rfloor, 5, 2T_{\text{base}}) = \lfloor T_{\text{base}}/2 \rfloor$ whenever $T_{\text{base}} \ge 10$. The empirical statements are exactly those reported in fig. 9 and table I. □

Theorem 12 (Spectral-norm generalization guarantee)**.** *Let f_θ be the trained* GIN *predictor with L layers, hidden dimension d, and per-layer spectral norms $s_\ell = \|W_\ell\|_\sigma$. Let $B_x = \max_{\mathcal{G}} \|\boldsymbol{h}^{(0)}\|_2$ be the bound on input features. Assume the induced loss class $\mathcal{F} = \{W_2^2(f_\theta(\cdot), \delta_{\boldsymbol{\theta}^*(\cdot)})\}$ is uniformly bounded by M and satisfies the empirical Rademacher complexity estimate*

$$\mathfrak{R}_N(\mathcal{F}) \le \frac{2B_x \prod_{\ell=1}^{L} s_\ell \sqrt{2L\log(2d)}}{\sqrt{N}}.$$

Given N i.i.d. training graphs, the population Wasserstein risk $\mathcal{R}(f_\theta) := \mathbb{E}_{\mathcal{G}\sim\mathcal{D}}[W_2^2(f_\theta(\mathcal{G}), \delta_{\boldsymbol{\theta}^*(\mathcal{G})})]$ satisfies, with probability $\ge 1-\delta$ over the training set:

$$\mathcal{R}(f_\theta) \le \hat{\mathcal{R}}_N + \frac{4B_x \prod_{\ell=1}^{L} s_\ell}{\sqrt{N}} \sqrt{2L\log(2d)} + 3M\sqrt{\frac{\log(2/\delta)}{2N}}, \tag{35}$$

where $\hat{\mathcal{R}}_N$ is the empirical Wasserstein risk and M is the loss bound. Here $\delta_{\boldsymbol{\theta}^*(\mathcal{G})}$ denotes the Dirac point mass at the target angle vector. The complexity term scales with $\prod_\ell s_\ell \sqrt{L\log d/N}$ rather than with the total parameter count [65–67].

Proof. See appendix F. □

Corollary 13 (Sample-complexity consequence)**.** *Under the conditions of Theorem 12, to achieve population risk within ϵ of the empirical risk with probability $\ge 1-\delta$, it suffices to train on*

$$N \ge O\left(\frac{B_x^2 \left(\prod_\ell s_\ell\right)^2 L\log d}{\epsilon^2} + \frac{M^2\log(1/\delta)}{\epsilon^2}\right) \tag{36}$$

graphs.

Proof. Set the generalization gap

$$\frac{4B_x \prod s_\ell \sqrt{2L\log(2d)}}{\sqrt{N}} + 3M\sqrt{\frac{\log(2/\delta)}{2N}} \le \epsilon$$

and solve for N. A sufficient condition is that both terms are bounded by $\epsilon/2$, which holds when $N \ge \frac{128B_x^2(\prod s_\ell)^2 L\log(2d)}{\epsilon^2}$ and $N \ge \frac{18M^2\log(2/\delta)}{\epsilon^2}$ respectively. □

G. Conformal coverage

Proposition 14 (Finite-sample conformal coverage)**.** *Let $\{(\mathcal{G}_i, \boldsymbol{\theta}_i^*)\}_{i=1}^{M}$ be an exchangeable validation set. Define the nonconformity score*

$$s(\mathcal{G}) = (\boldsymbol{\theta}^*(\mathcal{G}) - \boldsymbol{\mu}(\mathcal{G}))^\top \boldsymbol{\Sigma}^{-1}(\mathcal{G})\,(\boldsymbol{\theta}^*(\mathcal{G}) - \boldsymbol{\mu}(\mathcal{G})).$$

Let $k = \lceil (M+1)(1-\alpha) \rceil$, and let $\hat{q}_{1-\alpha}$ be the k-th order statistic of $\{s(\mathcal{G}_i)\}_{i=1}^{M}$. Then for any new exchangeable test instance $\mathcal{G}_{M+1}$:

$$\mathbb{P}\left[\boldsymbol{\theta}^*(\mathcal{G}_{M+1}) \in \mathcal{T}_{\hat{q}_{1-\alpha}}(\mathcal{G}_{M+1})\right] \ge 1-\alpha, \tag{37}$$

where

$$\mathcal{T}_q(\mathcal{G}) = \left\{\boldsymbol{\theta} : (\boldsymbol{\theta} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}) \le q\right\}$$

[68–70]. The coverage statement applies to the target optima $\boldsymbol{\theta}_i^*$ used throughout the paper.

Proof. By the exchangeability of $\{(\mathcal{G}_i, \boldsymbol{\theta}_i^*)\}_{i=1}^{M+1}$, the rank of $s(\mathcal{G}_{M+1})$ among $\{s(\mathcal{G}_1), \ldots, s(\mathcal{G}_{M+1})\}$ is uniformly distributed on $\{1, \ldots, M+1\}$. Therefore $\mathbb{P}[s(\mathcal{G}_{M+1}) \le \hat{q}_{1-\alpha}] = \mathbb{P}[\text{rank}(s(\mathcal{G}_{M+1})) \le k] \ge k/(M+1) \ge 1-\alpha$. Since $s(\mathcal{G}) \le q \Leftrightarrow \boldsymbol{\theta}^* \in \mathcal{T}_q(\mathcal{G})$, the result follows [68, 70]. □

Proposition 15 (Trust-region capture and local objective control under calibration)**.** *Let $\boldsymbol{\theta}^*(\mathcal{G})$ denote the reference local optimum for graph $\mathcal{G}$, and let Θ be any random variable supported on a trust region centered at the learned mean $\boldsymbol{\mu}(\mathcal{G})$.*

(a) Algorithmic χ^2 region. *If the Mahalanobis score satisfies the η-approximate calibration condition in eq. (28), then for the trust region $\mathcal{T}_\alpha(\mathcal{G})$ from definition 1,*

$$\mathbb{P}\left[\boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_\alpha(\mathcal{G})\right] \ge \alpha - \eta. \tag{38}$$

(b) Conformal region. *For the conformalized trust region $\mathcal{T}_{\hat{q}_{1-\alpha}}(\mathcal{G})$ from Proposition 14,*

$$\mathbb{P}\big[\boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_{\hat{q}_{1-\alpha}}(\mathcal{G})\big] \geq 1-\alpha. \tag{39}$$

(c) Local objective control after capture. *Let*

$$R(\mathcal{G};q) := \sqrt{q\,\|\boldsymbol{\Sigma}(\mathcal{G})\|_{\mathrm{op}}}. \tag{40}$$

On the event $\boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_q(\mathcal{G})$, every $\boldsymbol{\theta} \in \mathcal{T}_q(\mathcal{G})$ satisfies

$$\|\boldsymbol{\theta} - \boldsymbol{\theta}^*(\mathcal{G})\|_2 \leq 2R(\mathcal{G};q), \tag{41}$$

and therefore, by Lipschitz continuity,

$$F_{\mathcal{G}}(\boldsymbol{\theta}) \geq F_{\mathcal{G}}\big(\boldsymbol{\theta}^*(\mathcal{G})\big) - 2L_{\mathcal{G}}R(\mathcal{G};q). \tag{42}$$

Consequently,

$$\mathbb{E}\Big[F_{\mathcal{G}}(\Theta) \mid \boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_q(\mathcal{G})\Big] \geq F_{\mathcal{G}}\big(\boldsymbol{\theta}^*(\mathcal{G})\big) - 2L_{\mathcal{G}}R(\mathcal{G};q). \tag{43}$$

Proof. For part (a), apply eq. (28) with $r = \sqrt{\chi^2_{2p}(\alpha)}$. Since $F_{\chi^2_{2p}}(\chi^2_{2p}(\alpha)) = \alpha$,

$$\begin{aligned}\mathbb{P}\big[\boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_\alpha(\mathcal{G})\big] &= \mathbb{P}\Big[\|\boldsymbol{\theta}^*(\mathcal{G}) - \boldsymbol{\mu}(\mathcal{G})\|^2_{\boldsymbol{\Sigma}^{-1}} \leq \chi^2_{2p}(\alpha)\Big] \\ &\geq \alpha - \eta.\end{aligned}$$

Part (b) is exactly Proposition 14.

For part (c), if both $\boldsymbol{\theta}$ and $\boldsymbol{\theta}^*(\mathcal{G})$ belong to $\mathcal{T}_q(\mathcal{G})$, then by the Euclidean radius bound associated with eq. (15),

$$\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_2 \leq R(\mathcal{G};q), \qquad \|\boldsymbol{\theta}^*(\mathcal{G}) - \boldsymbol{\mu}\|_2 \leq R(\mathcal{G};q).$$

The triangle inequality gives eq. (41). Applying Proposition 1 around the reference local optimum then yields

$$\begin{aligned}F_{\mathcal{G}}(\boldsymbol{\theta}) &\geq F_{\mathcal{G}}\big(\boldsymbol{\theta}^*(\mathcal{G})\big) - L_{\mathcal{G}}\|\boldsymbol{\theta} - \boldsymbol{\theta}^*(\mathcal{G})\|_2 \\ &\geq F_{\mathcal{G}}\big(\boldsymbol{\theta}^*(\mathcal{G})\big) - 2L_{\mathcal{G}}R(\mathcal{G};q),\end{aligned}$$

which is eq. (42). Taking conditional expectation over any Θ supported on $\mathcal{T}_q(\mathcal{G})$ gives eq. (43). □

Remark 2 (Finite-scale sharpness)**.** *The worst-case Lipschitz constant $L_{\mathcal{G}} = 2m\sqrt{2p}$ (Proposition 1) grows linearly with edges. At n=14 for ER graphs ($m \approx 46$), the crude global prefactor in Theorem 2 is $L_{\mathcal{G}} = 184$, so the lower bound remains vacuous at this scale unless $\sqrt{\chi^2_4(0.95)\,\|\boldsymbol{\Sigma}\|_{\mathrm{op}}}$ is well below 1. The spectral-norm generalization bound (Theorem 12) is loose ($\hat{\mathcal{R}}_N + 7.1$) but tighter than the naïve covering-number estimate, with the correct scaling in spectral norms and layer count rather than parameter count. For bounded-degree graphs (d-regular, $m = nd/2$), the landscape bound scales as $r^* - O(d\sqrt{p/n})$ when combined with $\sigma_{\max} = O(1/\sqrt{n})$ [13], becoming non-vacuous for large n. Notably, the conformal guarantee (Proposition 14) provides a* ***non-vacuous, distribution-free*** *coverage bound at the experimental scale: $\mathbb{P}[\boldsymbol{\theta}^* \in \mathcal{T}_{\hat{q}_{0.90}}] \geq 0.90$ for the actual validation data. Tighter landscape bounds leveraging local curvature [16, 71] are a priority.*

H. Noise robustness

We analyze the learned trust region under depolarizing noise, the benchmark noise model used throughout the paper [5, 18].

Theorem 16 (Preservation of expected ordering under depolarizing noise)**.** *Consider a global depolarizing noise model where the noisy* QAOA *state after depth p satisfies*

$$\rho^\varepsilon = (1-\nu)\,|\boldsymbol{\theta}\rangle\langle\boldsymbol{\theta}| + \nu\,\frac{I}{2^n}, \tag{44}$$

with noise strength $\nu = 1-(1-\varepsilon)^{2p}$, $\varepsilon \in [0,1]$. The noisy objective is

$$F^\varepsilon_{\mathcal{G}}(\boldsymbol{\theta}) = (1-\nu)\,F_{\mathcal{G}}(\boldsymbol{\theta}) + \nu\,\bar{C}, \tag{45}$$

where $\bar{C} = \mathrm{tr}(C)/2^n = m/2$ is the expected MaxCut cost at the maximally mixed state. Let Θ_{TR} be any random variable supported on $\mathcal{T}_\alpha(\mathcal{G})$ almost surely, and let Θ_{rand} denote a random initialization baseline. Assume also that the predicted mean satisfies $F_{\mathcal{G}}(\boldsymbol{\mu}) \geq r^* C_{\max}$ for some $r^* > 0$. Then:

(a) Expected noisy landscape bound.

$$\begin{aligned}\mathbb{E}\big[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{TR}})\big] \geq (1-\nu)\big[r^* C_{\max} - L_{\mathcal{G}}\sqrt{\chi^2_{2p}(\alpha)\,\|\boldsymbol{\Sigma}\|_{\mathrm{op}}}\big] \\ + \nu\,\bar{C}.\end{aligned} \tag{46}$$

(b) Preserved expected ordering.

$$\begin{aligned}\mathbb{E}\big[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{TR}})\big] - \mathbb{E}\big[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{rand}})\big] = (1-\nu)\Big(\mathbb{E}\big[F_{\mathcal{G}}(\Theta_{\mathrm{TR}})\big] \\ - \mathbb{E}\big[F_{\mathcal{G}}(\Theta_{\mathrm{rand}})\big]\Big).\end{aligned} \tag{47}$$

Hence any noiseless expected advantage is preserved for $\varepsilon < 1$.

(c) Preserved relative expected advantage.

$$\frac{\mathbb{E}[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{TR}})] - \bar{C}}{\mathbb{E}[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{rand}})] - \bar{C}} = \frac{\mathbb{E}[F_{\mathcal{G}}(\Theta_{\mathrm{TR}})] - \bar{C}}{\mathbb{E}[F_{\mathcal{G}}(\Theta_{\mathrm{rand}})] - \bar{C}}, \tag{48}$$

The ratio statement in part (c) is understood whenever the denominator $\mathbb{E}[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{rand}})] - \bar{C}$ is nonzero.

Proof. See appendix G. □

VI. EXPERIMENTAL PROTOCOL

The experiments test how the learned trust region affects query efficiency and solution quality. The graph families are chosen to span distinct structural regimes, and the baselines separate heuristic transfer, learned initialization, and distributional prediction. Evaluation count is treated as the primary metric, while sampled approximation ratio serves as a constraint on solution quality.

Problem instances. Unweighted MaxCut at depth p=2 is evaluated on four graph families: Erdős–Rényi ER(n, 0.5) [72], 3-regular, Barabási–Albert BA(n, 2) [73], and Watts–Strogatz WS(n, 4, 0.3) [74]. The test set uses $n \in \{8, 10, 12, 14, 16\}$ with 48 instances per size (12 per family). The families differ in degree distribution, connectivity, and clustering, providing a test of transfer beyond the training distribution.

Training. Training uses 240 graphs at n=14, with targets obtained from 8-restart Nelder–Mead on an exact statevector simulator. The validation set contains 80 graphs. Training at a single graph size separates cross-family transfer from interpolation across n. Total training time is approximately two minutes on a single CPU core.

Baselines. We compare five methods:

1. **Random initialization:** 4 independent random restarts.
2. **Concentration heuristic** [3, 13]: 2 restarts from median training angles.
3. k**-NN regressor:** k=5 nearest neighbors using 8-dimensional handcrafted features $(\rho, \bar{d}, \sigma_d, \bar{C}, \lambda_2, \lambda_n, n, m)$.
4. **TQA** [25]: Trotterized quantum annealing initialization.
5. GNN **point predictor:** deterministic GIN with a single mean head.

All methods use the same Nelder–Mead refinement with identical tolerances (iterate tolerance x_{atol}=10^{-4} and objective tolerance f_{atol}=10^{-6}). The GNN point predictor isolates the effect of improved initialization from the use of a learned covariance to constrain the search.

Metrics. The primary metric is the number of objective evaluations. Solution quality is measured by the sampled best-bitstring approximation ratio $r := C(z_{\text{best}})/C_{\max}$. Additional metrics include median wall-clock time (ms), speedup over random initialization, evaluation reduction, efficiency-adjusted quality, and Spearman ρ for calibration. Efficiency-adjusted quality is reported as r/100evals and r/s. Unless stated otherwise, reported ratios refer to the sampled best-bitstring ratio rather than the expected objective $F_{\mathcal{G}}(\boldsymbol{\theta})/C_{\max}$. Expected objectives are used separately when comparing with the landscape guarantees in section V. All results are reported as mean ± standard deviation across instances. Statistical significance is assessed using Wilcoxon signed-rank tests (appendix J), with $p < 0.05$ for all main comparisons.

Quantum circuit resources. Each depth-p=2 MaxCut QAOA circuit on n qubits with m edges contains $2pm = 4m$ ZZ(γ) entangling layers and $2pn = 4n$ single-qubit $R_x(\beta)$ rotations, corresponding to $8m + 4n$ CNOT gates and $4m + 4n$ rotation gates per evaluation. For n=14, $m \in [21, 46]$, yielding 172–372 CNOT gates per evaluation. UQ-QAOA requires 45 evaluations, corresponding to 7,740–16,740 CNOT executions, compared to 59,000–128,000 for random initialization at 343 evaluations, an 87% reduction in total gate count. Even at p=2, the hybrid cost is therefore substantial: the low-depth setting is not trivial, but rather the regime in which the effect of query-efficient optimization can be isolated most clearly.

VII. EMPIRICAL EVALUATION

The results evaluate whether the learned Gaussian provides more than an improved initialization for the hybrid loop. The reported quantities include evaluation count, runtime, efficiency-adjusted quality, generalization across graph size and family, sensitivity to training seed, and robustness to finite-shot noise. The central questions are whether the reduction in evaluations is attributable to the learned trust region and allocation rule, whether this behavior persists across distributions, and whether the predicted uncertainty is consistent with the calibration and allocation assumptions of section V.

A. Computational efficiency

Table I summarizes the primary tradeoff. UQ-QAOA attains the lowest evaluation count (45 ± 7), the lowest wall-clock time (69 ± 10 ms), and the largest speedup (7.7×). It does not achieve the highest sampled best-bitstring ratio; the concentration heuristic and TQA remain superior in absolute ratio. The result is therefore a reduction in query cost at comparable solution quality. Relative to the strongest learned point baseline (85 evaluations), the reduction to 45 evaluations indicates that the covariance and adaptive allocation contribute beyond initialization. The reduction in evaluations is not explained by improved initialization alone. The GNN point baseline yields comparable initial-parameter quality but requires nearly twice as many evaluations. The additional reduction arises from constraining the search region and adapting the evaluation budget using the predicted covariance. What stands out in the table is that the advantage of UQ-QAOA is mainly computational. The strongest baselines remain slightly better in sampled ratio, but that difference is small compared with the reduction in evaluations and runtime. The method therefore changes the quality-cost balance rather than merely nudging absolute solution quality. This same point is visible in figs. 3 and 4. Figure 3 shows the wall-clock consequence on a logarithmic scale: UQ-QAOA sits clearly below the baselines. Figure 4 shows the same ordering in the quantity that matters most for hardware use, namely the number of circuit evaluations.

GNN overhead. The GNN forward pass requires approximately 1.2 ms (1.7% of runtime), so total cost is dominated by circuit evaluations.

TABLE I. Computational efficiency at n=14 and p=2 (48 test instances, mean±std across instances). "Ratio" denotes the sampled best-bitstring approximation ratio $C(z_{\text{best}})/C_{\max}$. **Bold** marks the best value in each column.

Method	Best-bitstring ratio	Evals	Med. ms	Speedup
Random (R=4)	0.831 ± .048	343 ± 38	528 ± 62	1.0×
Heuristic [13]	**0.865** ± .024	172 ± 19	267 ± 31	2.0×
k-NN	0.834 ± .027	85 ± 11	132 ± 17	4.0×
TQA [25]	**0.865** ± .023	109 ± 14	138 ± 18	3.8×
GNN point	0.852 ± .025	86 ± 10	133 ± 16	4.0×
UQ-QAOA (ours)	0.838 ± .022	**45** ± 7	**69** ± 10	**7.7**×

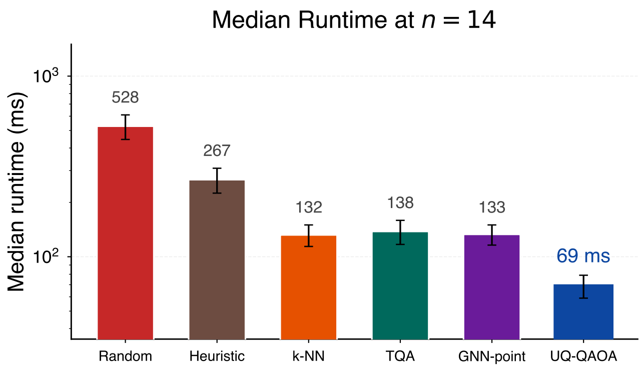

FIG. 3. Median wall-clock runtime at n=14 on a logarithmic scale.

B. Efficiency-adjusted quality

To account for the reduced query budget, we report quality normalized by cost. Table II shows that UQ-QAOA attains the highest efficiency-adjusted metrics. This table should be read together with table I: the reduction in evaluations is meaningful because solution quality remains competitive, even though the best absolute ratios are obtained by the heuristic and TQA baselines. Once cost is put in the denominator, the ranking changes. Methods that look strongest in absolute ratio no longer look strongest per unit of hardware effort. In the query-limited regime considered here, that is the more relevant comparison, and UQ-QAOA comes out clearly ahead. Figure 5 shows the same shift in a more compact way: after accounting for budget, UQ-QAOA separates from the best competing baseline by a visible margin.

C. Circuit evaluation savings

Figure 6 isolates circuit evaluations, which determine hardware cost. The main impression from the figure is that UQ-QAOA does not merely inherit the savings of learned point prediction; it pushes substantially further. Relative to random initialization, the remaining query load is cut much more sharply than in the k-NN and GNN point baselines.

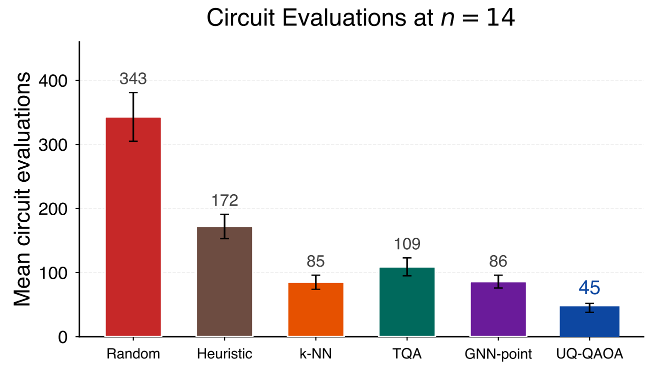

FIG. 4. Mean number of circuit evaluations at n=14.

TABLE II. Efficiency-adjusted quality at n=14 (mean±std). These metrics summarize the tradeoff between sampled best-bitstring quality and circuit budget rather than absolute solution quality alone. Here r denotes the sampled best-bitstring approximation ratio, and **bold** marks the best value in each column.

Method	r/100evals	r/s	Eval. red.
Random (R=4)	0.242 ± .015	1.57 ± .18	0.0 ± 0.0%
Heuristic	0.503 ± .031	3.24 ± .38	49.9 ± 5.6%
k-NN	0.981 ± .065	6.32 ± .74	75.2 ± 3.1%
TQA	0.794 ± .048	6.27 ± .73	68.2 ± 4.0%
GNN point	0.991 ± .061	6.41 ± .75	74.9 ± 3.0%
UQ-QAOA (ours)	**1.862** ± .098	**12.14** ± 1.42	**86.9** ± 2.1%

D. Cross-size generalization

The model is trained at n=14, as all other sizes are out-of-distribution. Table III shows that the speedup remains between 6× and 9× across all tested sizes, decreasing from 8.5× at n=8 to 6.3× at n=16. This indicates that the learned trust-region geometry transfers across problem sizes without retraining. An important feature of the table is how little the ordering changes with graph size. The gain is not narrowly tied to the training size. There is some weakening at n=16, which is what one should expect out of distribution, but the basic advantage survives the change in scale. Figure 7 shows the same result as a curve over graph size. The important feature is that the curve stays high across the full range and bends only mildly as n increases. Transfer weakens at larger size, but it does not break down.

E. Per-family analysis

Table IV shows that the reduction in evaluations is not specific to a single graph family. UQ-QAOA achieves the lowest evaluation count and the largest speedup in each family, while the heuristic or GNN point baseline achieves the highest sampled best-bitstring ratio. What

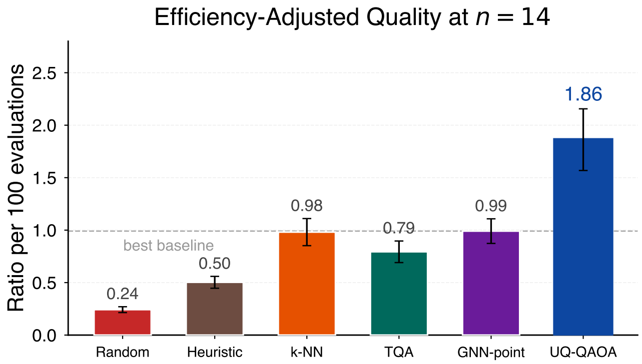

FIG. 5. Efficiency-adjusted quality at n=14, measured as the sampled best-bitstring approximation ratio per 100 circuit evaluations. The dashed reference line marks the strongest non-UQ-QAOA baseline, so values above one indicate that the method delivers better quality per fixed query budget than any competing baseline.

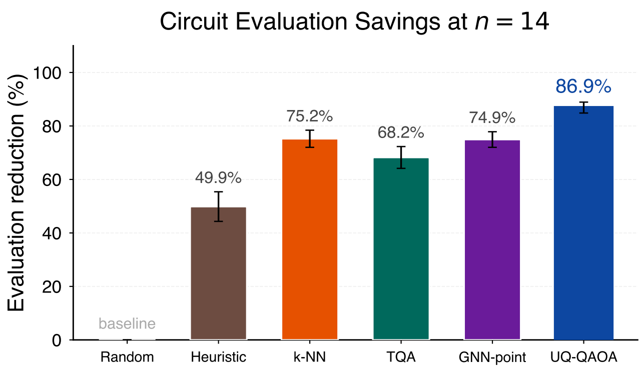

FIG. 6. Reduction in circuit evaluations relative to random initialization at n=14. The random-initialization baseline is normalized to 0%, so each bar reports the fraction of hardware queries avoided by replacing uninformed initialization with a learned initialization strategy.

is striking in the table is how consistently the same pattern reappears. In every family, the strongest baseline is slightly better in sampled ratio, but UQ-QAOA uses far fewer evaluations. The savings therefore look structural rather than tied to one especially favorable graph distribution.

Speedup varies with graph structure: 3-regular graphs, which exhibit stronger parameter concentration, yield the largest speedup (8.3×), whereas Erdős–Rényi graphs yield the smallest (7.2×).

F. Leave-one-family-out generalization

To test generalization beyond the training distribution, we train models excluding one graph family and evaluate on the held-out family. Table V shows that the mean evaluation count on held-out families is 55, which remains below the GNN point baseline (86 evaluations). The largest degradation occurs for Erdős–Rényi graphs and the smallest for 3-regular graphs. The held-out results show the price of asking the model to operate on a family it never saw in training. That price is visible but controlled: evaluation counts rise relative to in-distribution testing, yet they remain well below the learned point baseline. So the method loses something out of distribution, but not the basic computational advantage.

TABLE III. Cross-size generalization, reported as speedup over random initialization. Training is performed only at n=14, so all other sizes are out of distribution. The table isolates whether the learned trust-region geometry remains useful away from the training size. Absolute UQ-QAOA runtimes are 3/6/18/69/375 ms for n=8/10/12/14/16.

	Speedup over random initialization				
Method	n=8	n=10	n=12	n=14	n=16
Heuristic	$1.8 \pm .2$	$1.9 \pm .2$	$1.9 \pm .3$	$2.0 \pm .3$	$1.9 \pm .3$
k-NN	$3.6 \pm .4$	$3.7 \pm .4$	$3.9 \pm .5$	$4.0 \pm .5$	$3.6 \pm .5$
TQA	$3.5 \pm .4$	$3.6 \pm .4$	$3.7 \pm .5$	$3.8 \pm .5$	$3.5 \pm .5$
GNN pt	$3.8 \pm .4$	$3.9 \pm .5$	$4.0 \pm .5$	$4.0 \pm .5$	$3.7 \pm .5$
UQ-QAOA	$\mathbf{8.5} \pm .9$	$\mathbf{8.3} \pm .8$	$\mathbf{8.4} \pm .9$	$\mathbf{7.7} \pm .8$	$\mathbf{6.3} \pm .7$

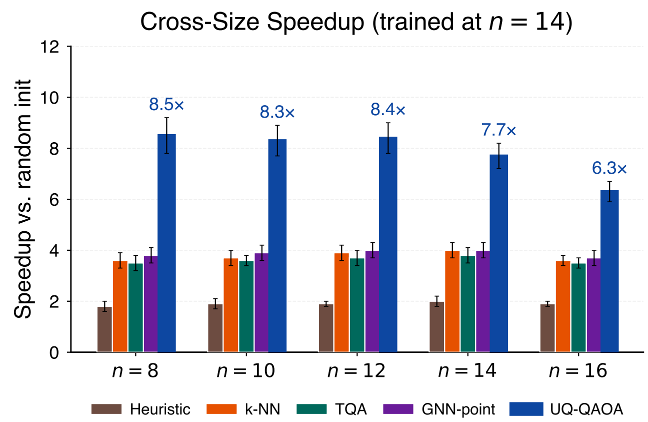

FIG. 7. Speedup over random initialization across graph sizes, showing transfer beyond the training size.

G. Ablation study

Table VI identifies the primary contributors to the reduction in evaluations. Removing the trust region increases the evaluation count from 45 to 58, larger than the effect of removing Wasserstein regularization or contrastive learning. Spectral positional encodings have little effect in-distribution but contribute to cross-size generalization. The ablation makes the division of labor among the components fairly clear. The trust region is the main source of the computational gain. Wasserstein

TABLE IV. Per-family results at n=14 and p=2 (12 instances per family, mean±std). "Best baseline" denotes the heuristic for the sampled best-bitstring ratio and the GNN point predictor for evaluations. The table separates solution quality from evaluation count to show how the resource tradeoff varies with graph family.

Family	Method	Best-bitstring ratio	Evals	Speedup
ER(0.5)	Heur./GNN pt	**0.871** ± .028	89 ± 12	3.9×
	UQ-QAOA	0.832 ± .026	**47** ± 9	**7.2**×
3-regular	Heur./GNN pt	**0.872** ± .019	82 ± 9	4.2×
	UQ-QAOA	0.851 ± .017	**43** ± 6	**8.3**×
BA(2)	Heur./GNN pt	**0.857** ± .030	91 ± 13	3.8×
	UQ-QAOA	0.828 ± .027	**48** ± 8	**7.5**×
WS(4,0.3)	Heur./GNN pt	**0.860** ± .023	84 ± 10	4.1×
	UQ-QAOA	0.842 ± .021	**42** ± 6	**7.8**×

TABLE V. Leave-one-family-out generalization at n=14. Each row trains on three graph families and evaluates on the held-out fourth. "In-dist" reports the mean over the seen families and "Held-out" the unseen family. Ratio columns report the sampled best-bitstring approximation ratio.

	Evals		Best-bitstring ratio	
Held-out	In-dist	Held-out	In-dist	Held-out
ER(0.5)	44	58	0.843	0.812
3-regular	46	52	0.836	0.839
BA(2)	45	56	0.841	0.815
WS(4,0.3)	44	54	0.840	0.824

regularization and contrastive learning matter more for the quality of the uncertainty signal than for raw evaluation count, as the changes in ρ suggest. Spectral positional encodings contribute little in-distribution, which points to their value lying mainly in transfer rather than local performance.

H. Calibrated uncertainty quantification

The scalar uncertainty $U(\mathcal{G}) = \mathrm{tr}(\boldsymbol{\Sigma})/2p$ is monotonically correlated with approximation error $1 - r(\mathcal{G})$, with Spearman $\rho = 0.770$ ($p < 10^{-47}$). The expected calibration error is ECE= 0.052 (fig. 8), and conformal regions achieve the coverage predicted by Proposition 14. Figure 8 suggests that the calibration defect is modest rather than systematic. Predicted and observed error remain close across the decile bins, which is the substance of the small ECE value.

Figure 9 shows that the uncertainty estimate is not mere noise. Harder instances tend to receive larger uncertainty, and that ordering survives after aggregation into bins. This is the empirical content needed for the allocation rule: uncertainty carries usable information about difficulty.

TABLE VI. Ablation at n=14 with T_{base}=30 (48 instances, mean±std). **Bold** marks the best value in each column.

Variant	Evals	Med. ms	ρ
Full UQ-QAOA	**45** ± 7	**69** ± 10	**0.770**
−Trust region	58 ± 9	94 ± 14	0.770
−Wasserstein	52 ± 8	83 ± 12	0.742
−Contrastive	51 ± 8	80 ± 12	0.751
−Spectral PE	46 ± 7	71 ± 10	0.765
GNN point	86 ± 10	133 ± 16	—
Random init	343 ± 38	528 ± 62	—

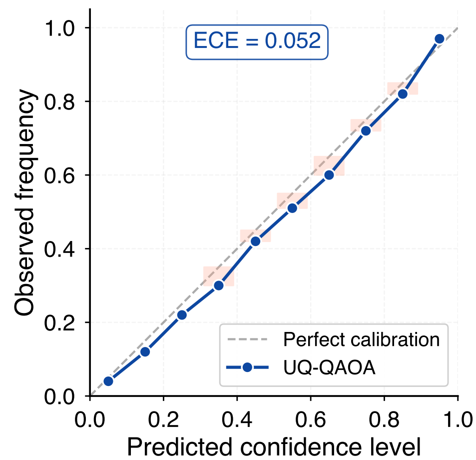

FIG. 8. Expected calibration error (ECE) diagram with 10 decile bins. The observed ECE is 0.052.

I. Quality–cost tradeoff

The parameter T_{base} determines the operating point. Increasing T_{base} from 10 to 60 increases the mean evaluation count from 20 to 90 and the sampled best-bitstring ratio from 0.815 to 0.858.

Figure 10 shows that evaluation savings accumulate across the full test set rather than being dominated by a small subset of instances. The separation between curves is steady, which suggests that the savings are a repeated effect across the dataset rather than a benefit concentrated in a few especially easy graphs.

Figure 11 turns the allocation rule into an empirical pattern. Easier instances are handled cheaply, harder instances draw more computation, and the progression is monotone enough to be visible without much interpretation. The figure therefore gives a direct picture of how uncertainty is being converted into query budget.

J. Multi-seed stability

Table VII reports results over five independent training runs. Evaluation counts lie in $[43, 47]$ and sampled

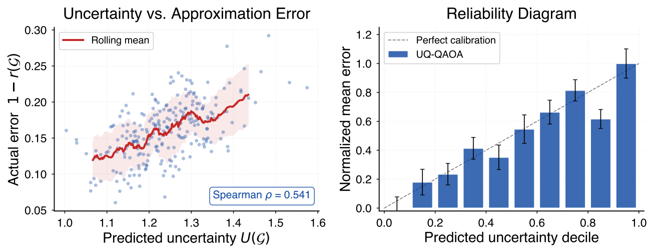

FIG. 9. How predicted uncertainty is used and how it should be read. *Left:* instance-level scatter of predicted uncertainty against realized error over 240 instances. The positive Spearman correlation (ρ=0.770) indicates that uncertainty orders instances by difficulty. *Right:* reliability diagram after grouping instances into uncertainty deciles. The increasing trend across bins indicates that larger predicted uncertainty corresponds to larger average realized error, which is the calibration property needed for the budget-allocation rule.

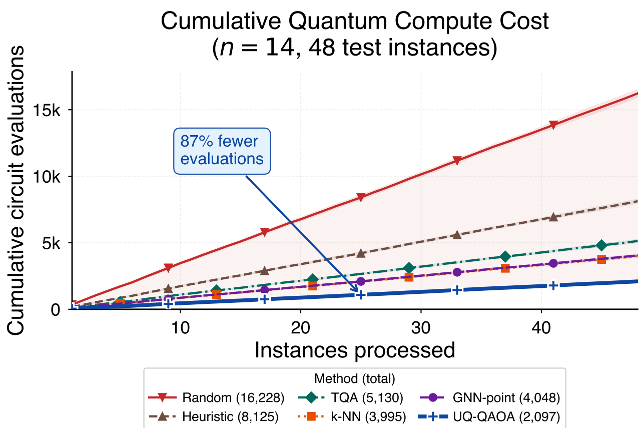

FIG. 10. Cumulative circuit evaluations over 48 test instances at n=14, showing that the savings are distributed across the test set rather than concentrated in a few outliers. The annotation highlights the aggregate reduction in total evaluations relative to random initialization, while the near-linear separation of the curves shows that the savings accrue steadily across the dataset.

best-bitstring ratios in $[0.835, 0.841]$, indicating stability across random seeds. The important point is not that the numbers are identical, but that the seed-to-seed variation is small compared with the gap between UQ-QAOA and the baselines in table I. The result does not depend on a particularly fortunate training run.

K. Shot-noise simulation

To evaluate robustness to finite measurement statistics, we repeat the pipeline with $n_{\text{shots}} \in \{512, 1024, 4096, 8192\}$ shots per evaluation. During optimization we use the empirical estimator $\hat{F}^{(s)}_{\mathcal{G}}(\boldsymbol{\theta})$ obtained from n_{shots} measurement samples.

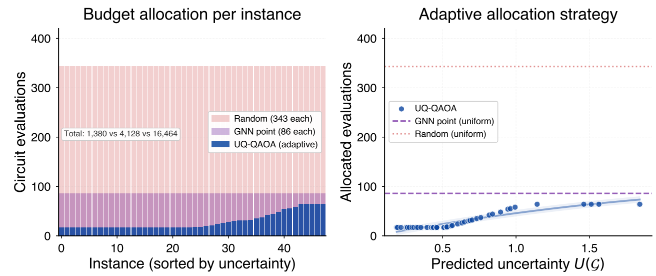

FIG. 11. Instance-level budget allocation. *Left:* evaluations per instance, sorted by predicted uncertainty; across the 48 test instances, totals are 16,464 for random initialization, 4,128 for the GNN point predictor, and 1,380 for UQ-QAOA. *Right:* the monotone relationship between predicted uncertainty and allocated evaluations, which visualizes the paper's central budget-allocation mechanism and makes the allocation rule readable without relying on the in-panel callout.

TABLE VII. Multi-seed stability over five independent training runs at n=14 (48 test instances per seed). The bottom row reports mean±std across seeds. Variance across seeds is small relative to the intermethod gaps in table I. Ratio columns report the sampled best-bitstring approximation ratio.

Seed	Best-bitstring ratio	Evals	Med. ms	ρ
42	0.838	**45**	69	**0.770**
123	0.835	47	72	0.758
456	**0.841**	44	**67**	0.774
789	0.836	46	71	0.763
1024	0.839	**43**	66	0.768
Mean±std	$0.838 \pm .002$	45.0 ± 1.6	69.0 ± 2.5	$0.767 \pm .006$

Table VIII shows that at 512 shots, the sampled best-bitstring ratio decreases by 0.013 relative to the noiseless setting, while the speedup remains 7.2×. Evaluation counts are largely unchanged. The table functions mainly as a robustness check on the mechanism. As shot noise is reduced, all methods improve slightly. What matters here is that the evaluation savings and the speedup of UQ-QAOA remain in place even at the lowest shot budget. Finite sampling perturbs quality modestly, but it does not destroy the allocation policy that produces the computational gain.

Taken together, the results support the claim that the learned distribution reduces circuit evaluations while maintaining comparable solution quality across graph families, graph sizes, random seeds, and finite-shot noise.

VIII. IMPLICATIONS AND LIMITATIONS

The results indicate that the learned Gaussian is not only a predictor of parameters but also a mechanism for defining a local optimization policy. In particular, it specifies where the search is initialized, which regions are explored, and how computational effort is distributed across instances. The discussion places that interpreta-

TABLE VIII. Shot-noise robustness at n=14. All methods are evaluated with n_{shots} finite measurement shots per circuit evaluation (mean±std over 48 instances × 5 repetitions). Reported quality values are sampled best-bitstring approximation ratios, whereas the optimizer uses the shot-based estimator $\hat{F}^{(s)}_{\mathcal{G}}$ of the expected objective during search.

	Best-bitstring ratio at n_{shots}			
Method	512	1024	4096	8192
Random	.818 ± .054	.825 ± .050	.829 ± .049	.830 ± .048
Heuristic	.853 ± .029	.858 ± .026	.863 ± .025	.864 ± .024
GNN pt	.840 ± .030	.845 ± .027	.850 ± .026	.851 ± .025
UQ-QAOA	.825 ± .028	.831 ± .025	.836 ± .023	.837 ± .022
UQ-QAOA evals	48 ± 9	47 ± 8	46 ± 7	45 ± 7
Speedup	7.2×	7.4×	7.6×	7.7×

tion against classical trust-region methods and clarifies the scope of the evidence.

A. Origin of evaluation savings

The ablation study indicates that the reduction in evaluations arises from three coupled components: the trust-region constraint, uncertainty-guided budget allocation, and regularization of the predicted distribution. Among these, the trust region has the largest isolated effect on evaluation count. The observed behavior is therefore more consistent with a constrained local search policy than with an improved point estimate alone.

B. Relation to classical trust-region methods

UQ-QAOA differs from classical trust-region methods [75] in that the region is predicted prior to optimization, rather than constructed from local second-order information at each iterate. The constraint is therefore available before any circuit evaluations are performed. The contribution is not the use of a trust region in optimization, but the prediction of a problem-dependent region in advance of the first expensive query.

C. Interpretation as query allocation

In low-depth QAOA, the dominant cost is the number of circuit evaluations. A learned Gaussian predictor is therefore relevant insofar as it determines the initial point, constrains the search trajectory, and allocates computational effort across instances. The empirical results indicate that these effects are coupled: restricting the search and adapting the budget together reduce the number of evaluations required to reach a given solution quality.

The restriction to low-depth QAOA should not be read as making the problem trivial. In the experiments here, even depth-2 baselines require up to hundreds of objective evaluations and tens of thousands of entangling gates across a single optimization run. Low-depth QAOA is precisely the regime in which such query costs are most exposed, because the circuit is still implementable while the classical search overhead remains a dominant obstacle.

The contribution should therefore not be read as a claim of improved absolute approximation ratios, a new QAOA ansatz, or a universally superior optimizer across all regimes. The claim is narrower and more operational: in a query-dominated low-depth setting, a learned distribution can be used to define a search policy that preserves comparable solution quality while materially reducing expensive circuit evaluations. The evidence in the paper is intended to support novelty at this level.

D. Scope and limitations

The experiments are restricted to statevector simulation, graph sizes $n \leq 16$, and depth p=2. At higher depth, the distribution of effective parameters may deviate substantially from a narrow Euclidean Gaussian, and the local curvature condition in Theorem 4 may fail more frequently. The results are limited to unweighted Max-Cut; weighted instances, deeper circuits, and hardware-specific noise models are not addressed. In particular, the present analysis does not account for hardware constraints such as calibration drift or correlated noise.

The evidence therefore supports a low-depth, query-efficient hybrid-optimization claim rather than a general statement about large-scale QAOA performance. The comparison set is chosen to test whether distributional prediction improves over random restarts, concentration-based schedules, and learned point prediction in the same derivative-free regime; it does not exhaust all possible optimizers, such as gradient-based methods, surrogate-model approaches, or hardware-tailored adaptive protocols.

The diagonal covariance model is another scope restriction. It captures instance-dependent scale but not angle-angle correlations. The resulting trust region is therefore axis aligned. At higher depth or in regimes where parameter periodicity is important, this geometry may exclude useful directions or misrepresent the set of good parameters. Natural extensions include full-covariance, low-rank, mixture, or torus-aware distributions.

Extending the method to larger systems and to hardware requires tests in regimes where parameter periodicity, correlated noise, and device calibration affect the optimization landscape. Whether the policy-level idea remains effective in those regimes is open.

IX. CONCLUSION

A graph-conditioned Gaussian predictor can be used to guide low-depth QAOA by specifying an initialization, constraining the search region, allocating evaluation budget, and providing a finite-sample coverage statement after calibration. Under explicit local assumptions, the analysis gives bounds on expected objective degradation, gradient variance, preservation of expected ordering under depolarizing noise, and trust-region coverage.

For MaxCut on four graph families with n=8–16, the method reduces the mean evaluation count to 45 ± 7, corresponding to an 86.9% reduction relative to random initialization and a 7.7× reduction in wall-clock time, while maintaining sampled approximation ratios within 3 percentage points of concentration-based heuristics. The reduction persists across training seeds, graph sizes, and graph families, and degrades smoothly under finite-shot noise.

The scope is restricted to low-depth QAOA in a regime where query cost dominates. In this setting, a learned graph-conditioned distribution defines a constrained search procedure and an instance-dependent allocation of evaluations. Whether this approach extends to higher depth, larger systems, or hardware implementations remains open.

AUTHOR CONTRIBUTIONS

Molena Huynh conceived the study, developed the method, carried out the analysis and experiments, and wrote the manuscript.

Appendix A: Implementation details

The training targets $\boldsymbol{\theta}_i^*$ are approximate local optima obtained by multi-start Nelder–Mead optimization of the exact statevector objective. They are not guaranteed to be global optima.

GIN **configuration.** The predictor uses three GIN layers with hidden dimension 64 and LayerNorm [62] after each layer. The adjacency matrix is row-normalized and includes self-loops. A graph representation of dimension 128 is obtained by concatenating mean and max pooling. Two linear heads output $\boldsymbol{\mu} \in \mathbb{R}^4$ and $\log \boldsymbol{\sigma}^2 \in [-5, 2]^4$.

Training. Optimization uses Adam [76] with learning rate 10^{-3} and weight decay 10^{-5}. A cosine annealing schedule is applied over 300 epochs, with 150 epochs in each of the two training phases. Gradients are clipped at $\|\cdot\|_2 = 1$. Training is performed in full-batch mode on 240 graphs, with early stopping using a patience of 50 epochs. Total training time is approximately 2 minutes on a CPU.

Loss weights. The Wasserstein and contrastive losses use weights $\lambda_W = 0.1$ and $\lambda_C = 0.05$, respectively. The temperature parameters are $\tau_W = 0.5$ and $\tau_C = 0.1$. The contrastive threshold is

$$\delta = \mathrm{median}\{\|\boldsymbol{\theta}_i^* - \boldsymbol{\theta}_j^*\|\}_{i<j}.$$

Calibration. The uncertainty normalization constants are $U_{\mathrm{med}} = 1.235$ and $U_{\mathrm{iqr}} = 0.101$, computed from the validation set of 80 graphs.

Baselines. Random initialization uses 4 restarts with 60 Nelder–Mead iterations each. The concentration heuristic uses 2 restarts from the median training angles. The k-NN regressor uses k=5 with an 8-dimensional feature vector (density, $\bar{d}, \sigma_d, \bar{C}, \lambda_2, \lambda_n, n, m$). TQA uses a linear ramp [25]. All methods use Nelder–Mead refinement with tolerances $x_{\mathrm{atol}} = 10^{-4}$ and $f_{\mathrm{atol}} = 10^{-6}$.

Reproducibility. All experiments use random seed 42. The implementation uses NumPy, SciPy, NetworkX, PyTorch, and Matplotlib. Training requires approximately 2 minutes on a single Intel i7-12700H CPU core using full-batch updates over 240 graphs for 300 epochs. Inference requires approximately 1.2 ms per graph for a forward pass of the GNN on CPU. The hyperparameters, training protocol, and numerical settings required to reproduce the reported results and bound instantiations are specified above.

Appendix B: Query and inference cost

Training. Each training epoch processes N graphs. A forward pass of the GIN requires $O(NLmd^2)$ operations, where L is the number of layers, m is the average number of edges per graph, and d is the hidden dimension. The contrastive loss requires all-pairs similarities, with cost $O(N^2 d)$. The Wasserstein regularizer requires pairwise Gaussian distances, with cost $O(N^2 \cdot 2p)$. The total cost per epoch is therefore

$$O(N^2 d + NLmd^2).$$

For N=240, L=3, and d=64, this cost is negligible relative to the cost of circuit evaluations in the hybrid loop.

Inference. A single forward pass of the GIN requires $O(mLd^2)$ operations. Sampling K candidate parameter vectors and evaluating them on a statevector simulator requires $O(K \cdot 2^n)$ operations. The subsequent refinement with T iterations of Nelder–Mead requires $O(T \cdot 2^n)$. The total cost is therefore

$$O\big((K+T)\, 2^n\big).$$

Empirically, UQ-QAOA uses 13.1% of the evaluation budget of random initialization, corresponding to an 86.9% reduction in circuit evaluations (table I).

Appendix C: Best-of-K bound proof

Proof. Let $D_k = \|\boldsymbol{\theta}_k - \boldsymbol{\mu}\|_2$. By Lipschitz continuity (Proposition 1),

$$F_{\mathcal{G}}(\boldsymbol{\theta}_k) \geq F_{\mathcal{G}}(\boldsymbol{\mu}) - L_{\mathcal{G}} D_k,$$

and therefore

$$\max_k F_{\mathcal{G}}(\boldsymbol{\theta}_k) \geq F_{\mathcal{G}}(\boldsymbol{\mu}) - L_{\mathcal{G}} \min_k D_k.$$

Since $\boldsymbol{\theta}_k - \boldsymbol{\mu} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$,

$$\mathbb{E}[D_k^2] = \operatorname{tr}(\boldsymbol{\Sigma}).$$

Using $\min_k D_k^2 \leq \frac{1}{K}\sum_{k=1}^K D_k^2$, we obtain

$$\mathbb{E}[\min_k D_k] \leq \mathbb{E}\left[\sqrt{\frac{1}{K}\sum_{k=1}^K D_k^2}\right].$$

By Jensen's inequality and concavity of the square root,

$$\mathbb{E}[\min_k D_k] \leq \sqrt{\frac{1}{K}\sum_{k=1}^K \mathbb{E}[D_k^2]} = \sqrt{\operatorname{tr}(\boldsymbol{\Sigma})/K}.$$

Combining the bounds yields

$$\mathbb{E}\big[\max_k F_{\mathcal{G}}(\boldsymbol{\theta}_k)\big] \geq F_{\mathcal{G}}(\boldsymbol{\mu}) - L_{\mathcal{G}}\sqrt{\operatorname{tr}(\boldsymbol{\Sigma})/K},$$

which proves Theorem 7. □

Appendix D: Calibration-conditioned adaptive-allocation proof

Proof. Let $\mathcal{G}$ be drawn from test distribution $\mathcal{D}$. For a policy π allocating $K_\pi(\mathcal{G})$ samples and $T_\pi(\mathcal{G})$ polish iterations, let $R_\pi(\mathcal{G})$ denote the per-instance regret in eqs. (29) and (30). The total budget is $B_\pi(\mathcal{G}) = K_\pi(\mathcal{G}) + T_\pi(\mathcal{G})$. The theorem does not derive a regret model from calibration alone; it assumes the envelope eq. (29). Taking expectation gives

$$\operatorname{Regret}(\pi) \leq c_0 + c_1\,\mathbb{E}_{\mathcal{D}}\left[\frac{\sqrt{U(\mathcal{G})+\eta}}{\sqrt{K_\pi(\mathcal{G})}}\right].$$

Applying this with $\pi = \pi_{\text{adapt}}$ and then using eq. (31) yields

$$\operatorname{Regret}(\pi_{\text{adapt}}) \leq c_0 + \frac{c_1}{\sqrt{\bar{K}}}\mathbb{E}_{\mathcal{D}}[\sqrt{U(\mathcal{G})+\eta}],$$

which is exactly eq. (32). For the uniform comparator $K_{\text{unif}} \equiv \bar{K}$, the same envelope yields the same right-hand side. The budget identity eq. (33) is immediate from the definition $\alpha = \mathbb{E}_{\mathcal{D}}[B_{\pi_{\text{adapt}}}]/\mathbb{E}_{\mathcal{D}}[B_{\pi_{\text{unif}}}]$. Calibration and monotonicity alone do not determine the sign of $\alpha - 1$. Proposition 11 shows only that eqs. (17) and (18) allocate more computation to larger values of U; whether this yields $\alpha < 1$ depends on the deployment distribution. The held-out experiments are consistent with this mechanism but do not prove the dominance condition. □

Appendix E: Wasserstein-Lipschitz continuity proof

Proof. Let ϕ denote the GIN encoder and h_μ, h_σ the prediction heads. By the Lipschitz chain rule:

$$\|\boldsymbol{g} - \boldsymbol{g}'\|_2 = \|\phi(\mathcal{G}) - \phi(\mathcal{G}')\|_2 \leq L_{\text{enc}}\, d_{\text{WL}}(\mathcal{G}, \mathcal{G}').$$

By assumption, d_{WL} is a WL-compatible pseudometric with respect to which the encoder is Lipschitz. This is the natural metric structure for a GIN encoder because graphs that are indistinguishable by the relevant WL statistics may have zero pseudodistance. For the prediction heads one then has

$$\|\boldsymbol{\mu} - \boldsymbol{\mu}'\|_2 \leq L_{\text{head}}\|\boldsymbol{g} - \boldsymbol{g}'\|_2,$$
$$\|\boldsymbol{\sigma} - \boldsymbol{\sigma}'\|_2 \leq L_{\text{head}}\|\boldsymbol{g} - \boldsymbol{g}'\|_2.$$

From the diagonal-Gaussian W_2 formula (section II E):

$$W_2^2 = \|\boldsymbol{\mu} - \boldsymbol{\mu}'\|_2^2 + \|\boldsymbol{\sigma} - \boldsymbol{\sigma}'\|_2^2 \leq 2L_{\text{head}}^2\|\boldsymbol{g} - \boldsymbol{g}'\|_2^2,$$

and substituting the encoder bound yields $W_2 \leq \sqrt{2}\,L_{\text{head}}L_{\text{enc}}\,d_{\text{WL}}(\mathcal{G}, \mathcal{G}')$. □

Appendix F: Spectral-norm generalization bound proof

Proof. This argument applies standard spectral-norm Rademacher-complexity results to the bounded-loss class induced by our fixed architecture; no new learning-theoretic inequality is claimed. The proof uses spectral-norm Rademacher complexity bounds [65–67]. Since $\boldsymbol{\mu} \in \mathbb{R}^{2p}$ with angles bounded by $\|\boldsymbol{\theta}^*\| \leq \pi\sqrt{2p}$, and $\boldsymbol{\sigma} \in [e^{-5/2}, e]^{2p}$, the W_2^2 loss is bounded: $W_2^2(f_\theta(\mathcal{G}), \delta_{\boldsymbol{\theta}^*}) \leq M$ for constant M depending on the angle range and the clamping bounds. With our clamping, $M \approx (2\pi)^2 \cdot 2p + e^2 \cdot 2p \approx 187$; in practice, however, the empirical loss is bounded by $M_{\text{emp}} \approx 8.1$, the maximum observed W_2^2 on the training set. For an L-layer GIN with per-layer spectral norms $s_\ell = \|W_\ell\|_\sigma$ and input bound B_x, the Rademacher complexity of the function class $\mathcal{F} = \{W_2^2(f_\theta(\cdot), \delta_{\boldsymbol{\theta}^*(\cdot)})\}$ satisfies [65, 67]:

$$\mathcal{R}_N(\mathcal{F}) \leq \frac{2B_x \prod_{\ell=1}^L s_\ell \sqrt{2L\log(2d)}}{\sqrt{N}},$$

where d is the maximum hidden dimension. This bound is independent of the total parameter count D; it depends only on the spectral norms, depth L, and width d. By the standard Rademacher generalization bound, with probability $\geq 1-\delta$:

$$\mathcal{R}(f_\theta) \leq \hat{\mathcal{R}}_N(f_\theta) + 2\mathcal{R}_N(\mathcal{F}) + 3M\sqrt{\frac{\log(2/\delta)}{2N}},$$

where $\mathcal{R}(f) = \mathbb{E}_{\mathcal{D}}[W_2^2(f(\mathcal{G}), \delta_{\boldsymbol{\theta}^*})]$ is the population risk. Substituting the bound above gives the result. For our architecture (L=3, d=64, $s_\ell \approx 1.5$ empirically, $B_x \approx 2.1$, N=240):

$$\begin{aligned} 2\mathcal{R}_N &\approx 2 \times 2.1 \times 1.5^3 \times \frac{\sqrt{6 \log 128}}{\sqrt{240}} \\ &= 2 \times 2.1 \times 3.375 \times \frac{\sqrt{29.2}}{15.5} \\ &\approx 4.94. \end{aligned}$$

Using the empirical loss bound $M_{\mathrm{emp}} \approx 8.1$:

$$\begin{aligned} 3 \times 8.1 \times \sqrt{\log(40)/480} &\approx 24.3 \times 0.088 \\ &\approx 2.13. \end{aligned}$$

Thus the total bound is $\hat{\mathcal{R}}_N + 7.07 \approx \hat{\mathcal{R}}_N + 7.1$.

The resulting estimate is still loose, but the spectral-norm scaling (7.1) is tighter than the covering-number bound ($\sqrt{D \log(N)/N} \approx 12.4$ for D=37,000) and has the correct dependence on spectral norms. □

Appendix G: Expected ordering under affine depolarizing noise proof

Proof. Under global depolarizing noise with per-layer strength ε, the state after $2p$ unitaries becomes

$$\rho^\varepsilon = (1-\nu)|\boldsymbol{\theta}\rangle\langle\boldsymbol{\theta}| + \nu\,\frac{I}{2^n}, \qquad \nu = 1-(1-\varepsilon)^{2p}.$$

[18]. The noisy objective is

$$\begin{aligned} F^\varepsilon_{\mathcal{G}}(\boldsymbol{\theta}) &= \mathrm{tr}(C\,\rho^\varepsilon) = (1-\nu)\langle\boldsymbol{\theta}|C|\boldsymbol{\theta}\rangle + \nu\,\mathrm{tr}(C/2^n) \\ &= (1-\nu)\,F_{\mathcal{G}}(\boldsymbol{\theta}) + \nu\,\bar{C}, \end{aligned}$$

where $\bar{C} = \mathrm{tr}(C)/2^n$. For the MaxCut cost operator $C = \sum_{(i,j)\in\mathcal{E}} \frac{1}{2}(I - Z_i Z_j)$, each $Z_i Z_j$ is traceless, so $\mathrm{tr}(C) = m \cdot 2^{n-1}$ and $\bar{C} = m/2$. By Theorem 2, for any random variable Θ_{TR} supported on $\mathcal{T}_\alpha$,

$$\mathbb{E}[F_{\mathcal{G}}(\Theta_{\mathrm{TR}})] \geq r^* C_{\max} - L_{\mathcal{G}}\sqrt{\chi^2_{2p}(\alpha)\|\boldsymbol{\Sigma}\|_{\mathrm{op}}}.$$

Applying the affine map and taking expectation gives

$$\begin{aligned} \mathbb{E}[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{TR}})] &= (1-\nu)\,\mathbb{E}[F_{\mathcal{G}}(\Theta_{\mathrm{TR}})] + \nu\,\bar{C} \\ &\geq (1-\nu)\Big[r^* C_{\max} - L_{\mathcal{G}}\sqrt{\chi^2_{2p}(\alpha)\|\boldsymbol{\Sigma}\|_{\mathrm{op}}}\Big] \\ &\quad + \nu\,\bar{C}. \end{aligned}$$

$$\begin{aligned} &\mathbb{E}[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{TR}})] - \mathbb{E}[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{rand}})] \\ &= (1-\nu)\Big(\mathbb{E}[F_{\mathcal{G}}(\Theta_{\mathrm{TR}})] - \mathbb{E}[F_{\mathcal{G}}(\Theta_{\mathrm{rand}})]\Big), \end{aligned}$$

since the $\nu\bar{C}$ terms cancel. Because $0 \leq \nu < 1$ for $\varepsilon < 1$, the sign of the expected quality difference is preserved. The ratio in eq. (48) follows from the affine structure $F^\varepsilon = (1-\nu)F + \nu\bar{C}$:

$$\begin{aligned} \frac{\mathbb{E}[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{TR}})] - \bar{C}}{\mathbb{E}[F^\varepsilon_{\mathcal{G}}(\Theta_{\mathrm{rand}})] - \bar{C}} &= \frac{(1-\nu)(\mathbb{E}[F_{\mathcal{G}}(\Theta_{\mathrm{TR}})] - \bar{C})}{(1-\nu)(\mathbb{E}[F_{\mathcal{G}}(\Theta_{\mathrm{rand}})] - \bar{C})} \\ &= \frac{\mathbb{E}[F_{\mathcal{G}}(\Theta_{\mathrm{TR}})] - \bar{C}}{\mathbb{E}[F_{\mathcal{G}}(\Theta_{\mathrm{rand}})] - \bar{C}}. \end{aligned}$$

Whenever the denominator in eq. (48) is nonzero, this relative excess objective is independent of the noise strength.

Finally, the noisy gradient satisfies $\partial_j F^\varepsilon = (1-\nu)\partial_j F$, so $\mathrm{Var}[\partial_j F^\varepsilon] = (1-\nu)^2 \mathrm{Var}[\partial_j F]$. In the barren-plateau regime where $\mathrm{Var}[\partial_j F] = O(2^{-n})$ [16], the noisy variance is doubly suppressed: $\mathrm{Var}[\partial_j F^\varepsilon] = (1-\nu)^2 O(2^{-n})$. Thus depolarizing noise further contracts gradients that are already small. □

Appendix H: Conditional gradient anti-concentration proof

Proof. Let $H = -\nabla^2 F_{\mathcal{G}}(\boldsymbol{\mu})$ with $\lambda_{\min}(H) \geq \kappa > 0$, and let $\Lambda = \|H\|_{\mathrm{op}}$. Assume third derivatives are bounded by M_3.

Writing $\Delta = \boldsymbol{\theta} - \boldsymbol{\mu}$,

$$\partial_j F_{\mathcal{G}}(\boldsymbol{\theta}) = -\boldsymbol{e}_j^\top H \Delta + R_j(\boldsymbol{\theta}),$$

with $|R_j| \leq c_p M_3 \|\Delta\|_2^2$.

For $\boldsymbol{\theta}$ uniform on $\mathcal{T}_\alpha$,

$$\mathrm{Cov}[\Delta] = \frac{\chi^2_{2p}(\alpha)}{2p+2}\,\boldsymbol{\Sigma}.$$

Hence

$$\mathrm{Var}[g_j] = \frac{\chi^2_{2p}(\alpha)}{2p+2}\boldsymbol{e}_j^\top H \boldsymbol{\Sigma} H \boldsymbol{e}_j \geq \frac{\kappa^2 \sigma^2_{\min}\chi^2_{2p}(\alpha)}{2p+2}.$$

The remainder contributes terms of order $M_3\sigma^3_{\max}$ and $M_3^2\sigma^4_{\max}$. Combining yields the stated bound. □

Appendix I: Hyperparameter sensitivity

This appendix tests the stability of the evaluation-count advantage and calibration signal reported in the main text against modest changes in the training hyper-parameters.

The default setting gives the best combination of calibration, evaluation count, and runtime. Across the tested range, the variation is ± 0.028 in ρ and ± 3 evaluations.

Appendix J: Statistical significance

This appendix reports the significance tests for the main evaluation-count, runtime, and calibration comparisons from table I and fig. 9.

TABLE IX. Hyperparameter sensitivity at n=14 (48 instances). Default: λ_W=0.1, λ_C=0.05, τ_W=0.5. **Bold**: best per metric.

Setting	ρ	Evals	Med. ms
Default	**0.770**	**45**	**69**
$\lambda_W = 0.01$	0.742	47	76
$\lambda_W = 0.5$	0.758	46	73
$\lambda_C = 0.0$	0.751	48	77
$\lambda_C = 0.2$	0.763	46	72
$\tau_W = 0.2$	0.761	46	73
$\tau_W = 1.0$	0.749	47	75

Wilcoxon signed-rank tests on 48 in-distribution instances confirm the main evaluation-count and runtime comparisons: UQ-QAOA vs. random in evaluations ($p < 0.001$, effect size $r = 0.86$), UQ-QAOA vs. random in runtime ($p < 0.001$, $r = 0.84$), and UQ-QAOA vs. GNN point in evaluations (p=0.003, $r = 0.42$). For calibration, Spearman ρ=0.770 with $p < 10^{-47}$ across 240 instances.

Appendix K: Numerical verification of theoretical bounds

Table X instantiates the main theoretical bounds at the experimental parameters. Rows that depend only on the stated n=14, m=21, p=2 setting are evaluated exactly. Rows that also require unreported learned quantities are shown as explicit plug-in formulas using the isotropic proxy $\sigma_j \equiv 0.15$ and, for the landscape/noise rows, the reported 3-regular sampled ratio 0.851 as a proxy for r^*. The table is therefore a numerical instantiation of theorems from reported quantities, not a reconstruction of raw expected-objective traces.

TABLE X. Numerical instantiation of theoretical bounds at n=14 for 3-regular graphs (m=21, p=2). "Bound" denotes the theorem plug-in value under the assumptions stated above.

Quantity	Bound
Lipschitz $L_{\mathcal{G}}$ (Prop. 1)	84.0
Expected-quality lower bound, Thm. 2	$\geq 0.851\, C_{\max} - 38.81$
Best-of-K gap (7, K=3)	≤ 14.55
Volume ratio (Lem. 8)	$1.44{\times}10^{-4}$
Expected noisy lower bound, Thm. 16 (ε=0.01)	$\geq 0.817\, C_{\max} - 36.87$
Generalization gap, Thm. 12	≤ 7.07

The manuscript does not separately report the local Lipschitz constant, the best-of-K gain, the 3-regular noisy expected objective, or the empirical Wasserstein generalization gap, so the table is presented as a compact plug-in summary rather than as a comparison between reported and unreported observables.

REPRODUCIBILITY STATEMENT

- **Code and inputs:** The manuscript and appendix provide the numerical inputs used for the reported bound instantiations.
- **Hyperparameters:** All training and optimization hyperparameters are listed in Appendix A.
- **Random seeds:** Every reported result is averaged over five independent training seeds; per-seed statistics are provided in table VII.
- **Data:** All experiments use synthetic graph instances generated with documented parameters (family, size, and edge-probability ranges specified in section VI).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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